

Blind Cost Derivations: Alaska Bulk Fuel Delivery by Mode

Independent open-web derivation of USD per gallon-mile transport rates
for truck, barge, plane, and ice-road fuel delivery

DOE MAS fuel-logistics routing model — cost-rate research
Four independent research agents, three-plus methods per mode,
with full derivation logs, source tables, and decision records

July 15, 2026

Blind Cost Derivations: Alaska Bulk Fuel Delivery by Mode

Prepared: July 15, 2026 (derivations + primary-source verification round) **Purpose:** Independent, open-web derivation of transport cost in **USD per gallon-mile (2026 dollars)** for each fuel-delivery mode in the DOE MAS fuel-logistics routing model: highway truck, marine barge, aircraft, and winter ice road / snow trail.

Methodology

Each mode was assigned to a separate research agent operating **blind**: no access to the model's current cost values, to the project's prior derivations, or to each other's work — so no estimate could anchor on an existing number. Each agent was required to:

1. Use at least **three genuinely independent methods** (published tariffs/contracts, engineering cost build-up, and revealed delivered-price differentials from state fuel-price surveys and contracts);
2. Keep a **derivation log** recording every search, source acceptance/rejection, and assumption made at each decision point;
3. Show all arithmetic and state all inflation/currency conversions;
4. Report a point estimate, range, confidence, and an explicit statement of what the figure includes and excludes.

Convergence between independent methods — not any single source — is the evidence standard.

Verification round (July 15, 2026). The original derivations were built from search-result snippets because direct page fetching was permission-blocked. After unblocking, a second round of agents fetched every chapter's flagged primary documents — the DCRA Winter 2026 price report (plus the state's ArcGIS community-price service), the State of Alaska heating-oil contract, the DEC/Econ One transport-cost filing, NOAA *Distances Between United States Ports*, the ISER 2010 report, the DCCED 2005 and DEC 2007 studies, the Everts/Wright Air/NAC tariff pages, the USAspending charter award, the UTC haul account, and the 267-page ASTAR trail study — and verified or corrected every load-bearing number in place. Each chapter now carries a dated verification addendum; snippet-era errors (a wrong hub baseline, two misattributed figures, a short route reconstruction, wrong source vintages) are corrected and logged. **All four point estimates survived verification**; the barge per-mile figures shifted ~10% on the NOAA-verified route distance.

Local-documents round (July 17, 2026). A third pass read manually downloaded primary documents the web round could not reach: the ATRI 2025 *Operational Costs of Trucking* report, the Econ One refining-industry study, the AEA **PCE FY2025 Statistical Report** (margin-free utility fuel prices by community), the State marine-diesel contract, the NSB FY26–27 budget book, the OMB CWAT project sheet, and the Northwest Arctic Borough **Noatak Winter Fuel Haul System** proposal. **All four point estimates held again.** Highlights: the last snippet-sourced ATRI figures were confirmed against the report's own tables; the plane chapter's assumed Fairbanks wholesale anchor was replaced with an empirical one from road-served PCE communities (\$3.60/gal); the ice-road traffic weighting is now data-backed (PCE gallons: Nuiqsut 279k vs. Atqasuk 275k — the assumed 50/50 regime split confirmed); the Noatak proposal confirmed Method A2's physics (3,000–5,000 gal/trip, 10–12 mph, \$425K tracked tractor) though it publishes no operating rate; and the PCE data delivered the report's cleanest single fact — **Sleetmute's utility pays \$4.49/gal vs. Bethel's \$4.45**, meaning ~310 river miles adds essentially zero realized freight and the village's ~\$5/gal retail premium is entirely storage/margin/working-capital.

Results

Mode	Point estimate (\$/gal-mi, 2026\$)	Range	Confidence	Basis of convergence
Road (highway tanker)	0.0007	0.00045–0.0011	Moderate	Econ One transport map (Valdez→Fairbanks \$0.20/gal truck — read off the source deck); ATRI build-up (Tables 8/11 verified: \$2.260 avg / \$2.32 specialized per mile); 11-point within-vendor state-contract distance regression (slope 0.00067, R² ≈ 0.75); USDOT BCA cross-check
Barge — ocean linehaul	0.00023	0.00017–0.00039	Med-high	ISER 2010 build-up (market rate \$0.19–0.22/gal verified p. 17); voyage-cycle build-up; NOAA-verified route (Anchorage→Bethel 1,109 nm = 1,276 sm); PCE FY2025 margin-free prices bound from above
Barge — river distribution leg	0.011	0.006–0.018	Med-high	DEC 2007 invoice-based rate (\$0.007/gal per map mile, verified p. 21); ISER small-barge cost (\$0.60/gal typical, verified p. 20); PCE Sleetmute=Bethel result confirms realized utility gradient is procurement-pooled, not marginal freight
Plane (bulk fuel by air)	0.025	0.015–0.042	Med-high	DCRA Jan 2026 revealed premiums (verified vs. state ArcGIS data); DC-6 build-up; ISER benchmark (verified p. 13); exact federal charter award (\$11,792.50 / 1,400 gal); PCE FY2025 two-part regression (fixed \$1.31/gal + \$0.0082/gal-mi at the efficient full-load end) with empirical hub anchor (\$3.60, Circle/Central); Wright Air & NAC tariffs corroborate Everts within 0–6%
Ice road / winter trail (single rate)	0.010	0.004–0.025	Moderate	Build-up calibrated to the verified 2026 Anaktuvuk haul and corroborated by the Noatak haul-system proposal (loads, speeds, \$425K tractor); within-borough DCRA differentials; regime traffic weights now data-backed (PCE gallons ≈ 50/50 Nuiqsut/Atqasuk); Ontario & NWT analogs; Tibbitt–Contwoyto economics

Cross-cutting findings

- 1. Fixed per-delivery costs dominate short hauls in every mode.** Every derivation independently found a two-part cost structure — a true per-mile rate plus a fixed per-gallon cost per delivery event (truck: ~\$0.20/gal distributor/drop fixed; barge: \$0.15–0.30/gal hub transfer plus ~\$1.50/gal village lightering; plane: ~\$2.20/gal per air delivery; ice road: load/unload and mobilization). A flat \$/gal-mi therefore systematically underprices short hops and overprices long hauls. Where the model keeps single per-mile rates, the point estimates above are calibrated to each mode's *typical* route length; the fixed components belong in intermodal transfer fees.
- 2. Mode ordering is economically coherent.** Ocean barge (0.00026) < road (0.0007) < ice road (0.010) < air (0.025) — each step roughly 3–40× the previous — which reproduces observed behavior: communities barge everything they can, truck what the road system reaches, build winter roads to displace

airlift, and fly only what they must.

3. The mid-2026 fuel price shock is a commodity effect, not a freight effect. The June 2026 (Iran/Hormuz) price spike moved delivered fuel prices by dollars per gallon, but the estimated freight pass-through is small (e.g., $\approx +\$0.02/\text{gal}$ on a barge voyage). All rates here are calibrated to pre-shock (Jan–early-2026) price relationships and are stated in real 2026 dollars.

4. Retail price differentials overstate freight. In every revealed-price method, the majority of a remote community's price premium proved to be storage, working capital, and retail margin rather than transport (e.g., $\sim 85\text{--}90\%$ of Bethel's $\$2.78/\text{gal}$ net premium is non-freight). The derivations use differentials only where the non-freight components could be netted out or bounded.

5. Ice road remains the widest-uncertainty mode. No published Alaska tariff or operating rate exists for winter fuel hauling — three rounds of verification confirmed this by exhausting the candidate sources (the 267-page ASTAR study contains no fuel-haul dollars; the UIC haul account publishes operations but no cost; the Noatak haul-system proposal specifies the full operation — 20–30 mi route, 3,000–5,000 gal/trip, ~ 200 trips/season, $\$425\text{K}$ tracked tractor — but prices nothing, deferring operating cost to "end-user pays"). The mode spans two operationally different regimes — engineered ice roads driven by conventional tankers ($\sim \$0.004/\text{gal-mi}$) and tundra snow trails worked by tracked cat-trains ($\sim \$0.02/\text{gal-mi}$) — and the regime traffic split is now data-backed at $\approx 50/50$ (PCE FY2025 gallons: Nuiqsut 279,232 vs. Atkasuk 274,877), so the single recommended rate ($\$0.010$) is a confirmed midpoint rather than a coin-flip, though it still misprices each regime by $\sim 2\times$ in opposite directions. The one remaining load-bearing assumption is the $\sim \$900/\text{hr}$ convoy rate; a real NSB/UIC contract figure (FOIA-able) would retire it.

Comparison to current model values

Mode	Model (friction_costs.py)	This derivation (verified)	Assessment
Road	0.000528	0.0007	Model is at the pure-carriage floor of the derived range; defensible, $\sim 25\%$ low vs. point
Barge	0.000556	0.00023 linehaul / ~ 0.0010 all-in at Bethel distance	Model value sits between linehaul-only and all-in — consistent once lightering is accounted; NOAA-verified route is 1,276 statute mi, not 1,800
Plane	0.033	0.025	Model $\sim 30\%$ above the verified point estimate; inside the range (upper half)
IceRoad	0.00667	0.010	Model $\sim 35\%$ below the verified point estimate; inside the range (lower half)

Report contents

Chapters 1–4 are the four agents' full reports: search strategy, derivation log (steps and decisions), methods with complete arithmetic, source tables with reliability grades, cross-method reconciliation, final estimates, and prioritized follow-ups. Each chapter closes with a **Verification addendum (2026-07-15)** logging exactly which primary documents were fetched, which numbers were confirmed (with page cites), and which were corrected.

Road (highway tanker) — \$/gallon-mile derivation

Tooling note (required disclosure): This chapter was originally drafted from **WebSearch result snippets** because WebFetch was permission-blocked. On **2026-07-15** the two primary state PDFs (OPPM heating-oil contract; DEC Fairbanks ULS/HS cost analysis) were fetched and read page-by-page, and the load-bearing numbers were verified or corrected in place (see the first Verification addendum). On **2026-07-17** a **second pass** was run against locally downloaded primary documents: the full **ATRI 2025 report PDF** (previously gated), the **Econ One** source deck behind the DEC transport figures, and the state **marine-diesel contract**. The ATRI headline and line-item figures are now verified against the primary tables; the Econ One transport figures' mode assignments were read directly off the source map; and the marine contract was found to contribute no road-system points (see the second Verification addendum at the end).

1. Search strategy

Three independent evidence tracks were pursued in parallel, mirroring the three required methods:

1. **Tariff / contract track** — queries for Alaska fuel-hauling rates in ¢/gallon, State of Alaska bulk-fuel procurement documents (Office of Procurement & Property Management statewide contracts, Online Public Notices bid tabulations), and regulatory cost filings (Alaska DEC ULSD transition cost analyses, which contain carrier-quoted route transport costs).
2. **Cost build-up track** — queries for ATRI *Operational Costs of Trucking* (2025 update, 2024 data), sector-level (specialized/tanker) \$/mile, Alaska-specific trucking rates and driver-wage premiums, and Alaska highway tanker payload capacities (Colville, Petro Star/Sourdough, Dalton Highway operations).
3. **Revealed-price track** — queries for the DCRA Alaska Fuel Price Survey (Winter 2026 report), AHFC fuel survey, ISER fuel-cost-components studies, and per-community heating-oil prices along the road system (Glennallen, Delta Junction, Tok, Valdez, Fairbanks).

Government PDFs and regulatory filings were prioritized over trade-press and broker marketing pages. Inflation adjustment uses BLS CPI-U.

2. Derivation log (steps and decisions)

1. **Initial fan-out (3 searches).** Found: (a) a State of Alaska public-notice attachment quoting "OPIS Anchorage No. 2 ULSD \$3.749" (bid tabulation — could not be opened); (b) ATRI 2025 update headline: industry average **\$2.260/truck-mile (2024)**; (c) the DCRA Fuel Price Survey landing pages and the **State of Alaska heating-oil master contract PDF** (~~08-heating-oil.pdf~~, last updated 2/13/2026) with per-community **OPIS differentials**: Glennallen \$0.3750–0.3825/gal, Delta Junction \$0.4511–0.4601/gal (Inlet Energy, priced 7/2/2025).
2. **Fetch attempts.** WebFetch denied on both state PDFs in the original session; on 2026-07-15 both PDFs were fetched and read in full (see Verification addendum).
3. **Contract deep-dive.** The contract PDF (Last Updated 6/30/2026) lists **FY26 and FY27 "Management Fees"** added to the OPIS Daily gross rack average (OPIS price location "LA" for road-system rows; Trimms is on OPIS ANCHORAGE, Juneau on SEATTLE, Southeast on PNW). The snippet-era ranges were the FY25–FY26 pair from the earlier 2/13/2026 revision; the fetched FY26 values confirm the upper ends: Wasilla \$0.2550 (Shoreside), Willow \$0.2383 (Inlet), Glennallen \$0.3825 (Inlet), Delta Junction \$0.4601 (Inlet), Fairbanks \$0.4048 (Crowley), Valdez \$0.6044 (Crowley), Two Rivers \$0.8484 (Crowley). **Corrected on verification:** Anchorage is \$0.2872 FY26 / \$0.2931 FY27 (Inlet Energy), not \$0.2718–\$0.2773 — the snippet figure was actually the Trimms row (Fisher's Fuel, FY26 \$0.2773). Also corrected: Willow and Anchorage share the *same* vendor (Inlet Energy, contract -H); Willow's lower fee reflects a flat "core zone" rate (\$0.2432 FY27 shared by Eagle River, Girdwood, Palmer, Moose Pass, Willow), not a vendor difference. **Decision:** use only *within-vendor, within-award* (Inlet Energy, 2022-0200-4928-H) points for the distance gradient. **Rejected:** Two Rivers (\$0.85 at ~25 mi from Fairbanks — clearly a small-volume delivery premium, not line-haul), Valdez (\$0.60 despite hosting a refinery — vendor/market artifact), Kenai Peninsula rows (served from the Nikiski/Kenai rack, so distance-from-Anchorage is the wrong regressor), and Eagle (\$1.53 — extreme remote Taylor Highway service).
4. **Regulatory filing found.** Alaska DEC's Fairbanks ULS/HS cost analysis (fetched and verified, p. 15, §1.3) states: "the rail costs between Anchorage and Fairbanks is approximately 16 cents/gallon ... This study also estimated the cost of transport for ULS fuel from PSI's Valdez refinery to Fairbanks at approximately 20 cents/gallon", plus ~2 ¢/gal Kenai→Anchorage. **Second-pass source read (2026-07-17):** DEC attributes these figures to Econ One's *The State of Alaska's Refining Industry*. The locally downloaded Econ One source is Barry Pulliam (Econ One Research, Inc.) presentation for the Alaska Oil & Gas Association, **Anchorage, Alaska, December 2016** — a slide deck, not the paginated 2015 report DEC cited by page number (pp. 10/25). Its slide "**Cost of In-State Product Movement Can Be Significant**" (p. 15) carries a mode-coded map with an explicit legend (Pipeline / Barge / Truck / Rail): **Valdez (Petro Star) → Fairbanks ≈ 20 ¢/gal is coded green = TRUCK** (confirming Method A's mode assumption directly, not merely by geographic necessity); **Anchorage → Fairbanks ≈ 16 ¢/gal is red = RAIL**; the **≈ 2 ¢/gal Kenai(Tesoro)→Anchorage figure is light-blue = PIPELINE** (not barge and not truck — **corrected**; earlier drafts called it barge, then "mode unstated"); a separate **≈ 10 ¢/gal movement is dark-blue = BARGE** (Kenai/Homer↔Southcentral). The companion slide (p. 14) shows import barge legs (PNW→Southcentral ≈ 25 ¢/gal; Asia ≈ 10 ¢/gal). **Vintage:** the deck is dated 2016 and carries no explicit dollar-year on the ¢/gal figures; because DEC cites a 2015 report and the earlier verification adopted a 2015 basis (CPI factor 1.409), the 2015→2026 adjustment is retained (using a 2016 basis instead would lower the factor only to ~1.375, moving Method A from 0.00077 to 0.00076 — within rounding, so the 0.00077 point is kept). The DEC text does not name modes; the Econ One map resolves them. **Decision (unchanged):** adopt the Valdez→Fairbanks truck figure as Method A's anchor (pure bulk refinery-to-rack line-haul). Rail figure kept only as a cross-modal sanity check; the 2 ¢ Kenai→Anchorage figure excluded (now confirmed pipeline, not truck).
5. **ISER studies located** (Components of Alaska Fuel Costs, 2010): "transportation costs range from a few cents per gallon between Valdez and Anchorage to over 40 ¢/gal for other routes"; confirms trucking is the cheapest rural mode. **Decision:** use ISER qualitatively only (2010 vintage, barge-focused); the DEC 2019 number supersedes it for trucking.

6. **Cost build-up inputs.** ATRI 2024 figures **verified against the primary PDF (2026-07-17)**, *An Analysis of the Operational Costs of Trucking: 2025 Update* (July 2025): industry average **\$2.260/truck-mile** (Table 8, p. 15); by sector (Table 11, p. 17) **Specialized \$2.32/mi**, Truckload \$2.13/mi, LTL \$2.49/mi — all confirmed to the cent. 2024 line-items (Table 8, p. 15): **fuel \$0.481**, truck/trailer payments \$0.390, **R&M \$0.198**, **truck insurance \$0.102**, tires \$0.047, permits/licenses \$0.009, tolls \$0.038; **driver wages \$0.798 + benefits \$0.197 = \$0.995/mi** combined comp. Per-hour (p. 14; Table 9, p. 16): **\$90.89 with fuel / \$71.57 without fuel** in 2024. **No tanker-specific line-item exists** — ATRI reports tank operations only inside the broad "Specialized" sector (its board includes J&M Tank Lines, but the report does not break tank truck out separately); Specialized total is \$2.32/mi and specialized driver comp runs **\$0.847–\$0.996/mi** by fleet size (Fig. 8, p. 25). Empty miles: non-tank trucks ran empty 16.7% nationally in 2024 (Fig. 21). **Cross-check:** USDOT BCA 2024 guidance uses a truck operating cost of **\$1.32/mi (2022 US\$)** at 45 mph / 7 mpg (POA Economic Assessment, pp. 83, 85) — a leaner "operating cost only" basis (excludes driver time, priced separately at \$33.50/hr) that brackets ATRI's fuel+equipment+R&M+insurance+tire subtotal (~\$0.83/mi) on the low side and the full \$2.26/mi on the high side. Alaska evidence: intra-/to-Alaska freight rates **\$2.80–\$3.10/mi** (trade sources); Alaska fuel-haul driver salaries \$95k–\$150k+ vs national median ≈ \$55k (FreightWaves). Tanker payloads in Alaska: Dalton Highway semis ≈ **10,000 gal**; Petro Star A-trains **13,000 gal**; Colville dual tankers 16,000 gal (North Slope-specific). **Decisions:** payload 10,500 gal central (9,000–13,000 range); deadhead factor 1.9 (fuel tankers return essentially empty; slight credit for occasional backhaul); Alaska rate \$3.00/mi central (market-rate basis, so carrier margin included).
7. **Revealed-retail attempt.** Per-community Winter 2026 DCRA / AHFC heating-oil prices for Tok/Glennallen/Delta could not be extracted from snippets (data live behind interactive maps/PDF tables). **Decision:** Method C pivots to the state-contract differential gradient (which is itself revealed delivered-price data at controlled quality), with the Fairbanks-vs-Delta cross-vendor spread as a secondary check. DCRA report used for context only.
8. **Inflation basis.** BLS CPI-U: 2015 annual = 237.017; 2019 annual = 255.657; 2024 annual = 313.689; June 2025 = 322.56 (BLS published values, from knowledge); June 2026 YoY = +3.5% (BLS release of 7/14/2026, via snippet) → June 2026 ≈ 322.56 × 1.035 = **333.9**. Factors: **2015→2026 = 1.409** (used for Method A after the vintage correction); 2019→2026 = 1.306; 2024→2026 = 1.064; Jul 2025→2026 = 1.035. Note that the FY27 contract fees are effective 7/1/2026 and therefore need **no** inflation adjustment.
9. **Distances** (standard Alaska highway mileposts, stated as assumptions): Anchorage→Willow 70 mi; →Talkeetna 113 (Parks MP 98.7 + 14-mi spur); →Chulitna ≈133 (assumed, Parks MP ~133); →Nelchina 137 (Glenn MP 137.5); →Glennallen 187; →Tazlina ≈191; →Gulkana ≈199; →Cantwell 210 (Parks MP); →Ernestine ≈240 (Richardson MP ~62 via Glennallen); →Healy 249 (Parks MP); Glennallen→Delta Junction 151 (Richardson MP 115→266), so Anchorage→Delta via Glennallen 338; Valdez→Fairbanks 364; North Pole rack→Delta Junction ≈ 95; Anchorage→Fairbanks (rail) 356; Fairbanks→Livengood ≈82, →Coldfoot ≈254, →Deadhorse ≈495 (Dalton corridor).

3. Method A / B / C — full derivations with arithmetic

Method A — Regulatory filing: bulk line-haul cost, Valdez refinery → Fairbanks (DEC report, citing Econ One)

Verified against the DEC PDF, p. 15 (§1.3): the 20 ¢ and 16 ¢ figures appear verbatim, attributed to Econ One. The Econ One source deck (Pulliam/AOGA, December 2016) was read on the second pass; its **map on p. 15 ("Cost of In-State Product Movement Can Be Significant")** codes the **Valdez→Fairbanks 20 ¢/gal movement as TRUCK (green)** and Anchorage→Fairbanks 16 ¢/gal as RAIL (red) via the legend — so the mode is now confirmed directly, not merely by geographic necessity. Vintage: DEC cites a 2015 Econ One report; the 2016 deck carries no explicit dollar-year, so the **2015→2026 CPI factor (1.409)** established on the first pass is retained (a 2016 basis would give ~1.375, immaterial at this rounding).

- Transport cost, Valdez refinery → Fairbanks (**truck, confirmed by Econ One's mode legend**): **\$0.20/gal** over **364 mi** (2015\$ basis).
- $\$0.20 \div 364 \text{ mi} = \mathbf{\$0.000549/\text{gal-mi (2015\$)}}$
- $2026\$: 0.000549 \times 1.409 = \mathbf{\$0.000774/\text{gal-mi} \approx \$0.00077/\text{gal-mi}}$
- Cross-modal sanity check: rail Anchorage→Fairbanks $\$0.16 \div 356 \text{ mi} = \0.000449 (2015\$) → $\$0.000633$ (2026\$). Rail correctly comes in below truck; truck/rail per-mile ratio 1.22 is plausible. (The Econ One map also shows a separate ≈ 2 ¢/gal **pipeline** Kenai→Anchorage and a ≈ 10 ¢/gal **barge** leg — neither is a truck movement, so both are excluded from Method A.)
- This is a refinery-to-rack bulk movement in ~10,000–13,000-gal units — the closest published analogue to a routing-model edge cost. Includes carrier operating cost + margin; excludes rack fees and distributor margin.

Method B — Cost build-up: Alaska tanker \$/mile ÷ payload

- National anchor (**ATRI 2024, verified primary tables**): Specialized sector **\$2.32/truck-mile** (Table 11, p. 17); industry average **\$2.260/mi** (Table 8, p. 15). Tank operations fall inside the Specialized sector — ATRI publishes no separate tanker line-item — so \$2.32 is the correct national starting point.
- Alaska uplift: driver comp ≈ 1.8–2× national (wages **\$0.798** + benefits **\$0.197 = \$0.995/mi** of the ATRI 2024 total, Table 8 → +\$0.80–1.00/mi at 1.8–2×); Alaska diesel above national average (fuel **\$0.481/mi** → +~\$0.08/mi); harsher R&M (\$0.198/mi base). Implied Alaska tanker cost ≈ \$2.95–\$3.25/mi. Independent market evidence: Alaska freight rates **\$2.80–\$3.10/mi** (rate basis, margin included); a broker page verified by fetch (heavyequipmenttransport.com) quotes Alaska truck services at **\$3.50–\$6.00/mi**, but that is heavy-equipment/oversize-oriented and skews high — treated as corroboration that Alaska rates sit well above the national \$2.26 average, not as a tanker rate. **Adopt \$3.00/truck-mile (2025–26\$, margin included).**
- Payload: **10,500 gal** central (Dalton semis ~10,000 gal; Petro Star A-trains 13,000 gal; conservative single-trailer 9,000 gal).
- Deadhead: fuel tankers return empty → round-trip factor **1.9** applied to the one-way delivery distance.
- Central: $(\$3.00 \times 1.9) \div 10,500 = 5.70 \div 10,500 = \mathbf{\$0.000543/\text{gal-mi} (\approx 2026\$)}$
- Low: $(\$2.80 \times 1.8) \div 13,000 = 5.04 \div 13,000 = \mathbf{\$0.000388}$
- High: $(\$3.25 \times 2.0) \div 9,000 = 6.50 \div 9,000 = \mathbf{\$0.000722}$
- **Method B: \$0.00054/gal-mi (range \$0.00039–\$0.00072), 2026\$.**

Method C — Revealed delivered-price gradient: State of Alaska heating-oil contract (OPIS + Management Fee)

Recomputed on verification (2026-07-15). The full contract table was fetched and read. Two upgrades over the snippet-era version: (1) fees are taken directly from the **FY27 column (effective 7/1/2026)**, so they are already in 2026\$ and need no CPI adjustment; (2) the regression now uses **eleven** within-vendor, within-award points (Inlet Energy, contract 2022-0200-4928-H, all on the same OPIS "LA" basis) instead of three, spanning both the Parks and Glenn/Richardson corridors out of the Anchorage rack.

Location	Distance (mi)	FY27 Management Fee (\$/gal)
Willow	70	0.2432
Talkeetna	113	0.3563
Chulitna	133	0.3959
Nelchina	137	0.3903
Glennallen	187	0.3903
Tazlina	191	0.4016
Gulkana	199	0.4016
Cantwell	210	0.4072
Ernestine	240	0.4355
Healy	249	0.4299
Delta Junction	338	0.4695

Excluded, with reasons: the flat "core zone" rows (Eagle River, Girdwood, Palmer, Moose Pass — all \$0.2432 regardless of distance, i.e., zone pricing carrying no distance signal; Willow shares this rate but is retained as the corridor's near anchor); Anchorage itself (\$0.2931 at 0 mi — self-delivery including local drayage); Kenai Peninsula rows (served from the Nikiski/Kenai rack); Big Delta/Birch Lake (\$0.5646 — small-drop premium next to Delta's \$0.4695); Nenana (plausibly Fairbanks-served); the Inlet -A award rows (Livengood/Manley/Central/Coldfoot/Deadhorse — Interior points served from the Fairbanks rack, used separately below); Eagle (\$1.5264 — extreme remote).

OLS on the 11 points: $\bar{x} = 187.91$, $\bar{y} = 0.392845$; $S_{xy} = 36.779$, $S_{xx} = 54,714.9 \rightarrow$ slope = $36.779 \div 54,714.9 = \text{\$0.000672/gal-mi (2026\$)}$, $R^2 \approx 0.75$ - intercept = $0.392845 - 0.000672 \times 187.91 = \text{\$0.267/gal}$ fixed component (distributor margin + delivery drop cost — a useful separate model parameter).

Specification sensitivity (all 2026\$): dropping Willow (treating the whole core zone as non-informative) $\rightarrow$ \$0.00044; adding Anchorage as a 0-mi anchor $\rightarrow$ \$0.00062; the original 3-point (Willow/Glennallen/Delta) regression on FY27 fees $\rightarrow$ \$0.00083 (essentially identical to the snippet-era \$0.000833, confirming that arithmetic). Pairwise: Willow $\rightarrow$ Glennallen $0.1471/117 = \$0.00126$; Glennallen $\rightarrow$ Delta $0.0792/151 = \$0.00052$.

Cross-vendor check assuming Delta is served from the Fairbanks/North Pole rack: (Delta 0.4695 – Fairbanks 0.4130) $\div$ 95 mi = **\$0.000595 $\approx$ \$0.00060 (2026\$)** — closely matching Methods A and B (previously \$0.000596 on FY25–FY26 midpoints; unchanged).

Dalton-corridor observation (Inlet Energy award -A, Fairbanks-rack supply): Livengood 0.4130 (~82 mi) $\rightarrow$ Coldfoot 0.6732 (~254 mi) $\rightarrow$ Deadhorse 0.9561 (~495 mi) gives pairwise slopes of \$0.00151 and \$0.00117/gal-mi — roughly **2 $\times$ the base gradient**, an empirical read on the Dalton Highway premium.

Method C: \$0.00067/gal-mi central (11-point OLS, 2026\$); specification band \$0.00044–\$0.00083; bulk-comparable cross-check \$0.00060. Biased high relative to pure line-haul because contract prices are delivered-to-tank (include final-mile service).

4. Source table

Source	URL	Year	What it provided
State of Alaska OPPM heating-oil master contract (08-heating-oil.pdf)	https://oppm.doa.alaska.gov/media/1337/08-heating-oil.pdf	Contract 2022-0200- 4928, term 1/1/2022– 6/30/2027; PDF last updated 6/30/2026	Fetched and verified 2026-07-15. Full Table A (73 mandatory locations) with FY26/FY27 Management Fees vs OPIS daily gross rack avg. Key FY26/FY27 values (\$/gal): Anchorage 0.2872/0.2931 (Inlet — corrected from snippet-era 0.2718–0.2773, which was the Trimms row), Wasilla 0.2550/0.2601 (Shoreside), Willow 0.2383/0.2432 (Inlet), Glennallen 0.3825/0.3903 (Inlet), Delta Junction 0.4601/0.4695 (Inlet), Fairbanks 0.4048/0.4130 (Crowley), Valdez 0.6044/0.6167 (Crowley), Two Rivers 0.8484/0.8656 (Crowley), Tok 0.4878/0.4977 (Shoreside), Coldfoot 0.6599/0.6732 (Inlet), Deadhorse 0.9371/0.9561 (Inlet), Eagle 1.4960/1.5264 (Inlet)

Source	URL	Year	What it provided
Alaska DEC, Fairbanks ULS/HS cost analysis (final draft)	https://dec.alaska.gov/media/16394/uls-hs-fairbanks-cost-analysis-finaldraft-05082019.pdf	Report Feb 2019/final 5/8/2019; transport figures cite Econ One	Verified 2026-07-15 (p. 15, §1.3): Valdez→Fairbanks ≈ 20¢/gal, rail ANC→FBX ≈ 16¢/gal, Kenai→ANC ≈ 2¢/gal. Modes confirmed 2026-07-17 against the Econ One source map
Econ One Research (Barry Pulliam / AOGA), <i>The State of Alaska's Refining Industry</i>	local: cost_derivation_resources/6_barry_pulliam-_econ_one_-_the_state_of_alaskas_refining_industry.pdf	Presentation, Anchorage, December 2016	Read 2026-07-17 . Slide p. 15 "Cost of In-State Product Movement" — mode-coded map (legend Pipeline/Barge/Truck/Rail): Valdez→Fairbanks 20¢ = truck ; ANC→FBX 16¢ = rail ; Kenai→ANC 2¢ = pipeline ; separate 10¢ = barge . Slide p. 14: import barge PNW→Southcentral 25¢, Asia 10¢. No explicit \$-year on the ¢/gal figures
ATRI, <i>An Analysis of the Operational Costs of Trucking: 2025 Update</i> (July 2025)	local: cost_derivation_resources/ATRI-Operational-Costs-of-Trucking-07-2025.pdf	2024 data	Read 2026-07-17 (no longer snippet-sourced) . Confirmed: industry avg \$2.260/mi (Table 8, p. 15); Specialized \$2.32/mi , Truckload \$2.13, LTL \$2.49 (Table 11, p. 17); 2024 line-items fuel \$0.481, R&M \$0.198, insurance \$0.102, wages \$0.798 + benefits \$0.197 (Table 8); per-hour \$90.89 w/ fuel, \$71.57 w/o (p. 14). No tanker-specific line-item (tank ops sit in "Specialized"). Empty 16.7% (Fig. 21)
POA (Port of Alaska) Economic Assessment (USDOT BCA)	local: cost_derivation_resources/POA Economic Assessment_202405013.pdf	Assumptions 2022 US\$; USDOT BCA 2024 guidance	Read 2026-07-17 : truck operating cost \$1.32/mi (2022\$) at 45 mph / 7 mpg, driver time \$33.50/hr (pp. 83, 85) — USDOT BCA baseline; used as a leaner operating-cost-only cross-check for Method B
FleetMaintenance / Cottingham & Butler ATRI summaries	https://www.fleetmaintenance.com/equipment/article/55301363/ ; https://www.cottinghambutler.com/post/2025-trucking-operational-costs-	2025	Sector \$/mi breakdown, wage/benefit components — now superseded by the primary ATRI PDF read 2026-07-17; no longer relied upon
State of Alaska OPPM marine-diesel & unleaded contract (08-marine-diesel-fuels.pdf)	local: cost_derivation_resources/08-marine-diesel-fuels.pdf ; https://oppm.doa.alaska.gov/media/1339/08-marine-diesel-fuels.pdf	Contract 2022-0200-4927; PDF last updated 6/30/2026	Read 2026-07-17 . All 26 rows are coastal marine (docksides/terminal) locations — Cordova, Dutch Harbor, Homer, Juneau, Ketchikan, King Salmon, Kodiak, Petersburg, Seward, Sitka, Skagway, Valdez, Whittier, Wrangell. No road-system inland rows , so it adds no within-vendor distance points to Method C
Heavy Equipment Transport, Alaska trucking page	https://www.heavyequipmenttransport.com/trucking/alaska.php	2025–26	Fetchd and verified 2026-07-15 : "The average cost of truck services in Alaska is approximately \$3.50 to \$6.00 per mile" — heavy-equipment oriented, skews high; used only as corroboration of the Alaska premium
Alaska Business Magazine, "Keeping Alaska Fueled Up" / Colville	https://www.akbizmag.com/industry/transportation/keeping-alaska-fueled-up/ ; https://www.colvilleinc.com/our-services/transport/	2021–2022	Tanker payloads: Dalton semis ~10,000 gal; Petro Star A-trains 13,000 gal; Colville duals 16,000 gal
FreightWaves, Alaska driver salaries	https://www.freightwaves.com/news/alaskan-trucking-fleets-promise-150k-driver-salaries-amid-drilling-boom	2024–2025	AK fuel-haul driver comp \$95k–\$150k+ vs ~\$55k national median
Alaska freight-rate trade pages (49th Freight,	https://www.49thfreight.com/ ; https://mercerc-trans.com/2026/02/09/shipping-freight-to-alaska/ ; https://usatruckerchoice.com/states/alaska/	2025–2026	AK trucking rates \$2.80–\$3.10/mi (and \$3.85–

Source	URL	Year	What it provided
Mercer, USA Trucker Choice)			4.50/mi spot for van/reefer/flatbed)
DCRA Alaska Fuel Price Report, Winter 2026	https://www.commerce.alaska.gov/web/Portals/4/pub/RA/Fuel_price_report/Alaska%20Fuel%20Price%20Report%20January%202026.pdf	2026	Context: survey window Jan 6–28, 2026; national heating-oil avg \$3.67; unsubsidized rural premium \$2.82/gal
ISER, Components of Alaska Fuel Costs	https://iseralaska.org/static/legacy_publication_links/componentsoffuel3.pdf	2010	Transport ranges "a few cents (Valdez–Anchorage) to >40¢/gal"; trucking cheapest mode
BLS CPI-U (June 2026 release; historical tables)	https://www.bls.gov/news.release/cpi.nr0.htm	2026	June 2026 CPI +3.5% YoY; historical annual averages (2019: 255.657; 2024: 313.689)
State bid tabulation w/ OPIS Anchorage ULSD \$3.749	https://aws.state.ak.us/OnlinePublicNotices/Notices/Attachment.aspx?id=92364	~2025	Not extractable — flagged for manual fetch

5. Cross-method comparison and reconciliation

Method	Basis	2026 \$/gal-mi	Bias assessment
A — DEC regulatory line-haul (Valdez→Fairbanks)	Bulk refinery→rack truck movement (mode confirmed via Econ One map), carrier-quoted	0.00077 (0.00055 in 2015\$)	Most directly on-target for an edge cost; single route, 2015 vintage inflated by CPI (trucking costs rose faster than CPI 2015–24, so possibly slightly low)
B — Cost build-up (ATRI + AK uplift ÷ payload)	Operating cost/market rate × deadhead ÷ gallons	0.00054 (0.00039–0.00072)	Sensitive to payload and deadhead assumptions; margin included via market-rate basis; ATRI inputs now verified against the primary tables (Specialized \$2.32/mi, line-items per Table 8), so the national anchor is firm — residual uncertainty is the Alaska uplift and payload, not the base data
C — State contract fee gradient (11-point, FY27)	Delivered-price vs distance, within vendor/award	0.00067 OLS (specification band 0.00044–0.00083); Fairbanks–Delta cross-check 0.00060	Includes final-mile small-drop service, not just line-haul; much better constrained after verification (11 points, R² ≈ 0.75)

The three methods bracket a band of roughly **\$0.0004–\$0.0011 per gallon-mile**, with the three central estimates (A = 0.00077, B = 0.00054, C = 0.00067) and the bulk-comparable cross-check (0.00060) clustering tightly around **\$0.0006–\$0.0008**. Verification *tightened* the cluster: A rose slightly (2015 vintage correction), and C's expanded regression fell from the knowingly-high 3-point value (0.00083) toward the middle of the band. Method B's low end assumes best-case 13,000-gal A-trains with efficient backhaul. The contract intercept (≈\$0.20–0.27/gal fixed, depending on specification) and the Two Rivers outlier (\$0.87/gal FY27 at ~25 mi) confirm that *fixed* per-delivery costs dominate short hauls — reinforcing that the per-mile slope, not the total differential, is the right routing-model quantity. The Dalton-corridor fees (Livengood→Coldfoot→Deadhorse, ≈\$0.0012–0.0015/gal-mi) bound the harsh-corridor upper end empirically.

6. Final estimate

Point value: \$0.0007 per gallon-mile (0.07¢/gal-mi, i.e., ≈ \$0.07/gallon per 100 road miles), 2026 dollars. Range: \$0.00045–\$0.0011 per gallon-mile. Use ~\$0.00045–0.0005 for high-volume A-train corridors with backhaul and ~\$0.0010+ for small-lot deliveries to minor road communities; the Dalton corridor prices out at ≈\$0.0012–0.0015 delivered. **Confidence: moderate-to-good** — two verification passes (state PDFs 2026-07-15; ATRI + Econ One + marine contract 2026-07-17) left the point value unchanged and tightened the cluster of central estimates to \$0.00054–\$0.00077 (A = 0.00077, B = 0.00054, C = 0.00067, cross-check = 0.00060). The **ATRI 2024 inputs to Method B are now verified against the primary tables**, and **Method A's truck-mode assumption is confirmed by the Econ One source map** — the two remaining snippet-sourced weaknesses from the first pass are now closed. Residual weakness: Method C's Anchorage-vs-Fairbanks supply-point ambiguity, and the Alaska driver-uplift multiplier in Method B (an assumption, not a measurement).

Includes: full highway tanker carriage — driver compensation, truck fuel, equipment, maintenance, insurance, carrier overhead and profit, and empty-return repositioning, expressed per gallon of delivered product per one-way route mile.

Excludes: the fuel commodity itself; terminal/rack loading fees; bulk storage; distributor fixed margin and final-mile drop service (empirically ≈ \$0.20–0.27/gal fixed in the state contract — recommend modeling this as a *per-gallon node/transfer cost*, not a per-mile cost); state/federal fuel taxes; winter/Dalton Highway surcharge conditions (the contract's Dalton-corridor fee gradient of ≈\$0.0012–0.0015/gal-mi implies roughly a **2× premium** over the base slope on that corridor — larger than the 20–50% previously guessed, though it includes small-drop service).

7. What would most improve this estimate

- 1. Remaining documents.** *Done on the second pass:* the ATRI 2025 report (line-items now verified; there is **no** tanker-specific breakout — tank ops sit inside "Specialized" at \$2.32/mi); the Econ One source deck (mode legend read — Valdez→Fairbanks 20¢ = truck, confirming Method A); the marine-diesel companion contract (all coastal — **no** road-system points, so no help to Method C). *Still open:* the **paginated Econ One 2015 report** DEC cited by page number (pp. 10/25) — the local file is the 2016 slide deck, which gives modes but not a stated dollar-year or payload context; the bid tabulation — <https://aws.state.ak.us/OnlinePublicNotices/Notices/Attachment.aspx?id=92364>; the DCRA Winter 2026 PDF and AHFC survey (per-community prices for a true retail-gradient regression).

2. **Resolve supply-point ambiguity** (Anchorage vs North Pole rack for Delta Junction and other Interior points) — still the largest swing factor in Method C's upper corridor, though the 11-point regression dilutes any single point's leverage.
3. **Obtain an actual Alaska fuel-carrier tariff or quote** (Colville, Petro Star, Crowley, Delta Western) for a named lane in ¢/gal — one confirmed lane rate would collapse the range substantially.
4. **Trucking-specific deflator**: replace CPI-U with an ATRI cost-index adjustment for the 2015→2026 escalation of the Method A anchor (trucking costs outpaced CPI over 2020–2023).

Verification addendum (2026-07-15)

WebFetch was enabled and the primary sources were fetched and read.

Fetched successfully: (1) the OPPM heating-oil contract PDF (all 4 pages, incl. the full 73-row Table A of mandatory locations and Table B remote locations); (2) the DEC Fairbanks ULS/HS cost analysis (all 31 pages + appendices); (3) the heavyequipmenttransport.com Alaska page. **Failed/gated:** the ATRI 2025 report landing page is a download form with no figures; the FleetMaintenance and Cottingham & Butler summary URLs and two candidate ATRI press-release URLs returned 404 — ATRI's \$2.260/mi and \$2.32/mi therefore remain snippet-sourced.

Confirmed: the contract FY26 Management Fees for Willow (0.2383), Glennallen (0.3825), Delta Junction (0.4601), Fairbanks (0.4048), Wasilla (0.2550), Two Rivers (0.8484), and Valdez (0.6044) — i.e., the upper ends of the snippet-era FY25–FY26 ranges (the current 6/30/2026 revision no longer shows FY25, so the lower ends are unconfirmable but consistent with the uniform ~2%/yr fee ladder); the DEC transport figures (20¢ Valdez→Fairbanks, 16¢ rail ANC→FBX, 2¢ Kenai→ANC) verbatim at p. 15, \$1.3; the \$3.50–\$6.00/mi Alaska trucking quote verbatim.

Corrected: (1) Anchorage's fee is **\$0.2872 FY26 / \$0.2931 FY27** (Inlet Energy), not 0.2718–0.2773 — the snippet had picked up the Trimms row; Anchorage was not used in any arithmetic, so nothing propagates. (2) The DEC transport figures are **2015-vintage** (Econ One Research 2015, pp. 10, 25), not 2019 — Method A's CPI factor becomes 1.409, moving Method A from \$0.00072 to **\$0.00077/gal-mi** and the rail cross-check from \$0.00059 to \$0.00063. (3) Willow and Anchorage are the *same* vendor; Willow's low fee is a flat core-zone rate, not a vendor effect.

Recomputed: Method C was rebuilt on the FY27 fees (effective 7/1/2026, so natively 2026\$) with 11 within-vendor Inlet Energy points across the Parks and Glenn/Richardson corridors: slope **\$0.00067/gal-mi**, intercept \$0.267/gal, $R^2 \approx 0.75$ (specification band \$0.00044–\$0.00083; the old 3-point spec reproduces at \$0.00083). New Dalton-corridor evidence (Livengood/Coldfoot/Deadhorse) prices that corridor at $\approx$ \$0.0012–0.0015/gal-mi.

Net effect on \$6: the final point estimate is **unchanged at \$0.0007/gal-mi (2026\$)**, with the central cluster now \$0.00054–\$0.00077 — verification tightened, rather than moved, the answer.

Verification addendum — second pass (local documents, 2026-07-17)

Four locally downloaded primary PDFs were read page-by-page, closing the two snippet-sourced items flagged after the first pass (ATRI figures; Econ One vintage/mode).

Documents read: (1) ATRI, *An Analysis of the Operational Costs of Trucking: 2025 Update* (July 2025) — full report; (2) Barry Pulliam / Econ One Research, *The State of Alaska's Refining Industry* (Anchorage, December 2016 slide deck) — the source behind DEC's transport figures; (3) State OPPM marine-diesel & unleaded contract 2022-0200-4927 (updated 6/30/2026); (4) POA Economic Assessment (USDOT BCA), read for the \$1.32/mi baseline.

Confirmed (ATRI, no longer snippet-sourced): industry average **\$2.260/truck-mile** (Table 8, p. 15) and Specialized **\$2.32/mi** (Table 11, p. 17) — both exact. 2024 line-items (Table 8, p. 15): fuel \$0.481, truck/trailer payments \$0.390, R&M \$0.198, insurance \$0.102, tires \$0.047, permits \$0.009, tolls \$0.038, driver wages \$0.798, benefits \$0.197. Per-hour (p. 14 / Table 9, p. 16): \$90.89 with fuel, \$71.57 without. **Finding:** ATRI has **no tanker-specific line-item** — tank truck operations are reported only within the broad "Specialized" sector (\$2.32/mi; specialized driver comp \$0.847–\$0.996/mi by fleet size, Fig. 8, p. 25). The earlier snippet-era "wages+benefits $\approx$ \$1.05/mi" was slightly high; the verified 2024 figure is **\$0.995/mi** — Method B's uplift arithmetic was refreshed accordingly (result unchanged).

Corrected / resolved (Econ One): the DEC transport figures trace to Econ One's mode-coded map (deck p. 15, "Cost of In-State Product Movement"). Reading the legend directly: **Valdez→Fairbanks 20¢/gal = TRUCK** (confirms Method A's mode assumption, previously justified only by "geographic necessity"); **Anchorage→Fairbanks 16¢/gal = RAIL**; the **2¢/gal Kenai→Anchorage figure = PIPELINE** (not barge, not truck — an outright correction of the earlier "mode unstated / previously barge" note); a separate **10¢/gal leg = BARGE**. The local file is the December 2016 deck (not the paginated 2015 report DEC cited by page); it carries no explicit dollar-year, so the 2015→2026 CPI factor (1.409) from the first pass is retained (a 2016 basis would be ~1.375 — immaterial). Method A stays at **\$0.00077/gal-mi**.

Marine-diesel contract: all 26 awarded rows are coastal marine (dockside/terminal) ports — Cordova, Dutch Harbor, Homer, Juneau, Ketchikan, King Salmon, Kodiak, Petersburg, Seward, Sitka, Skagway, Valdez, Whittier, Wrangell. **No road-system inland locations**, so it contributes **no new distance-gradient points to Method C** (as the task anticipated). Method C is unchanged (11-point OLS, \$0.00067/gal-mi).

New cross-check (POA / USDOT BCA): the USDOT BCA 2024 guidance truck operating cost of **\$1.32/mi (2022 US\$)** at 45 mph / 7 mpg with driver time priced separately at \$33.50/hr (POA Economic Assessment, pp. 83, 85) was added as a leaner operating-cost-only reference for Method B. It confirms \$1.32/mi as a plausible "hardware + fuel + driver-labor-excluded" operating floor well below ATRI's fully-loaded \$2.26/mi, consistent with the build-up.

Net effect on \$6: the final point estimate remains **\$0.0007/gal-mi (2026\$)**, range **\$0.00045–\$0.0011**. **Nothing moved** — the second pass converted two snippet-sourced inputs (ATRI figures; Econ One mode/vintage) into verified primary cites and corrected one factual detail (the 2¢ Kenai→Anchorage figure is pipeline, not barge), none of which feeds the arithmetic. Confidence rises from "moderate-to-good with two snippet caveats" to "moderate-to-good with the national anchor and the Method A mode both primary-verified."

Barge (marine linehaul + lightering) — \$/gallon-mile derivation

Method note: The original derivation (this file, first draft) was built entirely from WebSearch snippets because WebFetch was permission-blocked. On 2026-07-15 the primary sources were fetched and read directly (ISER 2010 local PDF; NOAA *Distances Between United States Ports* 2025; DCCED 2005; DEC/Northern Economics 2007; OPPM heating-oil contract sheet; DCRA Winter 2026). On 2026-07-17 a **second pass** added the two remaining primary documents (AEA PCE FY2025 Statistical Report by Community; state marine diesel contract 2022-0200-4927), introducing **Method E** — margin-free per-community utility fuel prices. Load-bearing numbers below are marked ✓ (verified, with page cites) or corrected in place; see the two **Verification addenda** at the end for the change logs.

1. Search strategy

Four independent evidence channels were pursued in parallel:

- 1. **Route distance** — NOAA *Distances Between United States Ports*, NOAA Coast Pilot, Port of Bethel/City of Bethel pages, Alaska Logistics port profiles, AMHS community pages; originally a leg-by-leg rhumb-line reconstruction (now superseded by the fetched NOAA tables — §3).
- 2. **Academic/government cost studies** — ISER/UAA *Components of Alaska Fuel Costs* (Feb 17, 2010) and *Components of Delivered Fuel Prices in Alaska* (2008), DCCED *Current Community Conditions: Fuel Prices Across Alaska* (Dec 2005), ADEC/Northern Economics *Cost Assessment for Diesel Fuel Transition in Western and Northern Alaska Communities* (Dec 2007), AEA PCE statistical reports, Alaska DOT Statewide Freight Assessment.
- 3. **Revealed price differentials** — DCRA Alaska Fuel Price Report (Winter 2026), OPPM state heating-oil contract 2022-0200-4928, AAA/Stacker Anchorage pump prices, KYUK / Alaska Beacon / Northern Journal 2026 reporting on Bethel, Sleetmute, Mountain Village, Togiak prices and the mid-2026 Iran-war price shock.
- 4. **Operational build-up inputs** — carrier fleet descriptions (Crowley, Vitus, Bowhead, Alaska Logistics), tug day-rate and fuel-burn literature, voyage-cycle parameters (season length, trip counts, barge capacities from ISER and carrier brochures).

Roughly 17 distinct search queries were executed in the original pass; six primary documents were fetched and read in the verification pass.

2. Derivation log (steps and decisions)

Rows annotated [V] = verified against the fetched primary source; [C] = corrected on verification (details in §§3–6 and the addendum).

#	Action	Finding / decision
1	Searched Anchorage→Bethel barge distance	No published through-distance found <i>in snippets</i> . [C] One exists: NOAA Tables 33+35 chain Anchorage→Unimak Pass→Bethel (§3).
2	Searched ISER fuel-cost studies	Found the two canonical ISER reports (2008, 2010) and research summary RS_68.
3	Searched KYUK Bethel fuel prices	Confirmed a mid-2026 commodity price shock (Iran war): Bethel gas \$6.70→\$9.30–9.37/gal June 2026. Decision: use pre-shock (summer-2025-priced) figures for freight inference; treat 2026 spike as commodity, not freight.
4	Attempted WebFetch of ISER PDF	Blocked in original session. [V] Local copy read 2026-07-15 (pp. 12–25).
5	Queried NOAA distances/Coast Pilot	[C] NOAA <i>distances.pdf</i> (2025, 14th ed.) fetched: Table 33 Gulf of Alaska + Table 35 Bering Sea list Anchorage, Unimak Pass, False Pass, Dutch Harbor, Naknek, Dillingham, Platinum, Bethel , Nome, etc. — supersedes the rhumb-line reconstruction (§3).
6	Queried ISER for barge cost build-up	[V] ISER pp. 15–18: linehaul barges 2.5–3.5M gal; set fixed cost \$2.5M/season (p. 16); tug fuel + misc "another \$3.5 million" (p. 17); 3M gal x 6 trips = 18M gal; "approximately \$0.19 per gallon ... transportation only" (p. 17); hubs = Dillingham, Naknek, Bethel, Nome, Kotzebue (p. 17); Bethel 12-ft controlling depth , 18–20-ft-draft linehauls lighter off one-third of cargo (pp. 17–18). Ambiguity resolved in §6. [C] The "distribution adds about \$1/gal" reading was wrong: \$0.98 is the total distribution cost incl. linehaul (Table 4, p. 22); the village small-barge leg alone is \$0.60 typical, \$0.40–0.80 range (Tables 2–3, p. 20).
7	Queried DCRA Winter 2026 report + Anchorage benchmarks	Bethel unleaded \$6.72/gal (winter 2025–26 survey); Anchorage \$3.56/gal (AAA, June 6, 2025). [V/C] DCRA PDF fetched: it is summary-only (per-community cells live in the online Community Database). It confirms statewide avg \$6.63, Western region avg gasoline \$7.97 / heating fuel \$7.85 , and the summer-price lock-in mechanism; the individual Bethel/village cells rest on KYUK/Beacon reporting of the survey.
8	Queried village-level prices	Sleetmute (far-upriver): \$9.43 pre-shock vs Bethel \$6.72; Mountain Village \$8.71; Togiak \$7.56 (secondary sources citing the state survey; consistent with DCRA regional averages). Bethel→Sleetmute ≈ 310 river mi.
9	Queried 2005 DCCED Community Fuel Report	[C] Fetched: the DCCED 2005 report contains no barge freight rates — it is a retail price survey. The "\$0.13–0.15/gal (2000–2005)" linehaul figure is actually in ISER 2010, p. 17 . DCCED 2005 contributes: 83% of 100 surveyed communities barge-served (p. 4, Table 4) and Nov-2005 retail anchors (Bethel gas \$3.61, Sleetmute \$5.25, Mountain Village \$4.40, Togiak \$4.51 — appendix).
10	Queried DEC/Northern Economics ULSD report (2007)	[C] Fetched: the report does not contain the \$0.19–0.22 ocean-linehaul figure (that is ISER 2010 p. 17, misattributed by a snippet). It does contribute a direct, invoice-based river rate: "about \$0.007 per gallon per map mile of transport from the fuel hub ... a village 100 map miles upriver ≈ 70¢ per gallon" (p. 21, MAFA analysis of Crowley invoices), plus barge sizes 63k–5.9M gal (Table 7, p. 21), volume discounts (Table 8, p. 23), FY2006 PCE utility prices by region (Table 10, p. 24), air delivery 1.0–1.4¢/gal per map mile (p. 73).
11	Queried PCE FY2024 and AVEC reports	Report located but per-community numbers not extractable from snippets. Rejected for arithmetic; listed for manual fetch. AVEC/Vitus context retained.
12	Queried tug day rates 2024–25	No clean public day-rate. [V] ISER p. 17: \$3.5M/120 useful days ≈ \$30,000/day (2010\$) for the linehaul's variable+misc; small coastal set \$10,000–12,857/day (Tables 2–3, p. 20).
13	Queried tug fuel burn	3,000–5,000 hp ocean tugs burn ≈3,000–5,000 gal/day towing. Adopted 3,200 gal/day loaded / 2,500 light.
14	Checked RS_68 snapshot	Bethel \$4.58 vs Anchorage \$4.25 (June 2008), differential \$0.33 — a stale-inventory timing artifact; "1,800 miles" rejected as a distance source.
15	Quantified the 2026 shock	Crowley quoted Yupiit School District +\$2.50/gal YoY (spring 2026) — attributed to commodity; marine-fuel pass-through ≈ \$0.02/gal (§4).

#	Action	Finding / decision
16	Route reconstruction	[C] Original 10-leg rhumb-line build gave Nikiski→Bethel ≈ 935 nm — ~10% short . NOAA-verified: Anchorage→Bethel = 1,109 nm (§3).
17	Inflation factors	BLS CPI-U annual averages (assumption): 2005=195.3, 2007=207.3, 2010=218.1, 2025=322; mid-2026 ≈330. Factors: 2005→2026 ×1.69 ; 2007→2026 ×1.59 ; 2010→2026 ×1.51 . (ISER figures now escalated from 2010, the report's date, not 2009.)

3. Route distance determination — ✓ NOAA-verified

Primary source: NOAA, *Distances Between United States Ports*, 2025 (14th) edition; distances in nautical miles. Table 33 (Gulf of Alaska Distances, p. 48) and Table 35 (Bering Sea and Arctic Ocean Distances, p. 50) share the junction point **Unimak Pass**; Bethel (60°49.0'N, 161°47.0'W — the city, i.e., including the river transit) is a listed port.

Verified legs (nm):

Leg	NOAA distance	Table
Anchorage → Unimak Pass	688	33
Anchorage → False Pass	636	33
Anchorage → Kodiak	242	33
Homer → Unimak Pass	573	33
Unimak Pass → Dutch Harbor	75	33/35
Unimak Pass → Bethel	421	35
Unimak Pass → Naknek / Dillingham / Platinum	377 / 384 / 307	35
Unimak Pass → Nome	646	35
Dutch Harbor → Bethel	464	35
Naknek → Bethel / Dillingham → Bethel / Platinum → Bethel	315 / 322 / 120	35
Bethel → Nome	620	35

Through-distances (derived from verified legs):

Route	Distance
Anchorage → Bethel via Unimak Pass	688 + 421 = 1,109 nm ≈ 1,276 statute miles
Nikiski → Bethel (Nikiski ≈ 45 nm down-inlet of Anchorage)	≈ 1,064 nm ≈ 1,224 sm
False Pass (Isanotski) shortcut for shallow-draft tows	saves ≈ 50–70 nm (Anchorage→False Pass 636 vs →Unimak 688; Bering-side rejoin)
Ocean-only (Anchorage → Kuskokwim river bar; Bethel is ~75 nm upriver)	≈ 1,034 nm ≈ 1,190 sm
Anchorage → Dutch Harbor (cross-check)	688 + 75 = 763 nm ≈ 878 sm — consistent with AMHS "about 900 miles southwest of Anchorage"

The original rhumb-line reconstruction (935 nm Nikiski→Bethel) was **~10% short**, almost entirely on the Gulf-of-Alaska legs; its Bering-side estimate (False Pass→Bethel ≈ 379 nm) agrees well with the NOAA-implied ≈370–420 nm. River mileage (~75 nm bar→Bethel; 12-ft controlling depth ✓ ISER pp. 17–18) unchanged.

Canonical distances used below: 1,250 sm linehaul (Anchorage/Nikiski→Bethel; range 1,150–1,300), of which ~86 sm is river. Five-hub rotation (Naknek 1,065, Dillingham 1,072, Bethel 1,109, Nome 1,334, Kotzebue ≈1,590 nm via Deering proxy): mean ≈ 1,234 nm ≈ 1,420 sm.

4. Method A / B / C (+D) — full derivations

Common deflator assumption: CPI-U, mid-2026 ≈ 330 (2005: ×1.69, 2007: ×1.59, 2010: ×1.51, 2025: ×1.025).

Method A — ISER 2010 engineering build-up (Cook Inlet → Bering Sea hubs) ✓ verified

- ✓ ISER (Feb 2010), pp. 16–17: tug+barge "set" fixed cost **\$2.5M/season**; tug fuel + misc for a 6–7-trip Cook Inlet↔Bering Sea season "**another \$3.5 million**"; 3M gal/barge × 6 trips = 18M gal/season; "approximately **\$0.19 per gallon** in cost for transportation only."
- Ambiguity resolved (§6):** the text describes a \$6.0M total season cost (→ \$0.33/gal full cost), but the report's own arithmetic and its market-rate observation ("current **\$0.19–\$0.22 per gal**," rising "**to over \$0.25**" with double-hull replacement — p. 17 ✓) establish \$0.19–0.22 as the *charged* rate. Carry **\$0.19–0.22 (2010\$) as the market rate** and **\$0.33 as the full-economic-cost ceiling**.
- 2026\$: market rate 0.19–0.22 × 1.51 = **\$0.29–0.33/gal**; full-cost ceiling 0.33 × 1.51 = **\$0.50/gal**.
- Distance basis: Bethel route 1,276 sm (NOAA ✓); five-hub rotation mean ≈ 1,420 sm.
- Per-mile: 0.29/1,276 = 2.3×10^{−4} \$/gal-mi** (hub-mean basis 2.0×10^{−4}; full-cost ceiling 3.9×10^{−4}).

Method B — Published/quoted rate history (all from ISER 2010 p. 17 ✓; DCCED reattributed)

- ✓ Rate history (ISER p. 17): late 1990s <**\$0.09/gal** (single-hull surplus, below cost — one such carrier went bankrupt); 2000–2005 **\$0.13–0.15/gal**; 2010 **\$0.19–0.22/gal**; projected >**\$0.25** as OPA-90 double-hull replacement completes (a cost that has since fully materialized — 2026 rates should sit at or above the escalated 2010 rate).

- *Correction:* the \$0.13–0.15 figure was previously attributed to DCCED (2005); the fetched DCCED report is a retail survey with no freight rates. Its contribution is context: **83% of 100 surveyed communities are barge-served** (Table 4, p. 4 ✓) and Nov-2005 retail anchors (Bethel gas \$3.61 ✓).
- *Correction:* the ADEC/Northern Economics (2007) report also contains no ocean-linehaul rate; its contributions moved to Method D (river rate) and Method C context (barge fleet specs, Table 7 ✓).
- 2026\$: 2005-era rate $\times 1.69 = \mathbf{\$0.22\text{--}0.25}$; 2010 rate $\times 1.51 = \mathbf{\$0.29\text{--}0.33}$; ">\$0.25 near-future" $\times 1.51 = \geq \mathbf{\$0.38}$.
- Over 1,276 sm: $\mathbf{1.7 \times 10^{-4} - 3.0 \times 10^{-4} \text{ per gal-mi}}$, central $\approx 2.3 \times 10^{-4}$.

Method C — Independent voyage-cycle cost build-up (2026 inputs, NOAA distances)

- Voyage: 1,064 nm (Nikiski) loaded at 8 kt = 5.5 d; return light at 9.5 kt = 4.7 d; load/discharge/bar-lightering = 2.5 d $\Rightarrow$ **12.7 d/round trip**.
- Fixed cost: ISER \$2.5M/season (2010\$ ✓ p. 16) $\times 1.51 = \$3.78\text{M}$ / 165-day season = \$22.9k/day $\times$ 12.7 d = **\$291k/trip** (cross-check: ISER's own variable metric $\approx$ \$30k/day 2010\$ ✓ p. 17 $\approx$ \$45k/day 2026\$ for fuel+misc-inclusive daily rate — same order).
- Tug fuel: 5.5 d $\times$ 3,200 gal/d + 4.7 d $\times$ 2,500 + 2.5 d $\times$ 1,000 $\approx$ **31,850 gal/trip**; at \$3.75/gal $\approx$ **\$119k** (at mid-2026 shock \$5.25/gal: \$167k).
- Cargo delivered: 3M-gal barge draft-limited at Bethel (12-ft controlling depth, lighter one-third ✓ ISER pp. 17–18) $\Rightarrow$ **2.0–2.5M gal/trip**.
- Cost/gal: (291k+119k)/2.5M = **\$0.16/gal** to (291k+167k)/2.0M = **\$0.23/gal**; +40% overhead/insurance/margin $\Rightarrow$ **\$0.23–0.32/gal** $\Rightarrow$ /1,224 sm = **$1.9 \times 10^{-4} - 2.6 \times 10^{-4}$ per gal-mi**.

Method D — Revealed price differentials (bounds + distribution leg)

- **Bethel vs Anchorage (matched commodity basis):** Bethel gas \$6.72 (winter 25/26 survey per KYUK/Beacon; DCRA PDF is summary-only but shows Western region avg \$7.97 ✓ and statewide \$6.63 ✓, consistent) – Anchorage \$3.56 (June 2025) = **\$3.16/gal all-in adder**; less Bethel 6% sales tax ($\approx$ \$0.38) $\Rightarrow \approx$ **\$2.78/gal** covering freight + tank-farm storage + 12-month working capital + rural retail margin. With A–C freight at \$0.2–0.4, ~85–90% of the revealed adder is *non-freight* — freight $\leq$ \$0.5/gal is an upper bound. (June-2008 ISER snapshot: Bethel–Anchorage = \$0.33 — in rising markets the differential compresses toward pure freight, bracketing from below.)
- **OPPM state heating-oil contract 2022-0200-4928 (sheet dated 6/12/2026 ✓):** Bethel (Vitus Energy) \$7.4970 less-tax retail – \$0.2800 discount = **\$7.2170/gal**. Anchorage is a Table A "standard delivery" location priced at **OPIS LA daily gross rack + \$0.2872 mgmt fee** — no posted flat price, so an exact same-contract Bethel–Anchorage differential requires the OPIS rack print for the delivery date. Same-sheet marine-served comparators: Dutch Harbor \$3.5500, Kodiak \$2.7600, Aniak \$6.2220, Dillingham \$7.549, Naknek \$7.949, St. Mary's \$9.3931. Bethel–Dutch Harbor = **\$3.67** and Bethel–Kodiak = **\$4.46** — an order of magnitude above any freight difference; and **Aniak, ~100 river mi beyond Bethel, is \$1.00/gal cheaper than Bethel** on the same sheet. Inventory-timing and local storage economics dominate revealed differentials — **do not** calibrate transport edges to them. (ISER Table 4 decomposition ✓ p. 22 makes the same point structurally: of a \$0.98 total distribution cost, linehaul is only \$0.19.)
- **Village distribution (lightering/small-barge) leg — two verified anchors:**
- ✓ ISER Tables 2–3 + text (p. 20): typical coastal small-barge delivery **\$0.60/gal**, difficult rivers \$0.77, stated range **\$0.40–0.80/gal** (2010\$) $\Rightarrow \times 1.51 = \mathbf{\$0.91\text{--}1.21 \text{ range} \text{ (2026$)}}$. (*Corrects the earlier "\$1.00/gal distribution" reading — that was Table 4's total \$0.98 including linehaul.*)
- ✓ ADEC/Northern Economics (2007), p. 21: river fuel barge $\approx$ **\$0.007/gal per map mile** from the hub (Crowley invoice analysis; 100 mi $\approx$ \$0.70) $\Rightarrow \times 1.59 = \mathbf{1.1 \times 10^{-2} \text{ $/gal-mi (2026$)}}$; at 82 map mi this reproduces the ISER \$0.91 typical figure almost exactly.
- Revealed check: Sleetmute \$9.43 – Bethel \$6.72 = \$2.71/gal over $\approx$ 310 river mi (includes village retail margin) $\Rightarrow \leq 8.7 \times 10^{-3}$ \$/gal-mi — consistent, and confirms per-mile rates fall on long legs (per-stop fixed costs dominate).

Commodity-shock separation (mid-2026): Crowley's +\$2.50/gal YoY quote and Bethel's \$6.70→\$9.30 jump are commodity effects. Freight pass-through of costlier tug fuel $\approx 31,850 \text{ gal} \times \$1.50 / 2.25\text{M gal} \approx \mathbf{\$0.02\text{/gal} \approx +7\% \text{ on freight}}$ — freight per-mile rates are essentially shock-invariant. (ISER corroborates the fixed-cost structure: a \$1.00/gal barge-fuel increase moves a small set's annual cost only ~5% — p. 19 ✓.)

Method E — PCE FY2025 utility fuel prices (margin-free revealed-freight cross-section) ✓ NEW, primary-source

Source (✓ fetched & read, 2026-07-17): AEA, *Power Cost Equalization Program Statistical Report by Community, FY2025* (final, dated 3/1/2026; reporting period **July 1 2024 – June 30 2025**). Each community page reports the utility's "**Average Price of Fuel**" = total fuel cost ÷ gallons used — the delivered bulk price paid by the generating utility, **net of any retail/heating-oil-dealer margin, sales tax, or working-capital carry on stored retail inventory**. This is exactly the "margin-free" revealed price §8 flagged as the top remaining refinement. The dollar-year is **FY2025 fuel purchases — largely pre-shock** (the mid-2026 Iran-war commodity spike falls entirely *after* this window), so these are clean 2025\$ figures.

Extracted per-community fuel prices (\$/gal, FY2025):

Community	Utility	\$/gal	Gallons	Route from Cook Inlet (sm)	River mi past Bethel	Class
Kotzebue	Kotzebue Electric Assn	3.33	1,197,401	$\approx$ 1,830 (1,590 nm)	—	ocean hub
Dillingham	Nushagak Electric	3.54	1,215,976	1,233 (1,072 nm)	—	ocean hub
Nome	Nome Joint Utility	3.59	2,026,320	1,534 (1,334 nm)	—	ocean hub
Naknek	Naknek Electric	3.86	1,515,424	1,225 (1,065 nm)	—	ocean hub
St. Mary's	AVEC	4.06	301,661	$\approx$ 1,300 (Yukon)	—	Yukon
Akiachak	Akiachak Native Comm	4.43	145,953	$\approx$ 1,294	$\approx$ 18	Kusko river
Bethel	AVEC	4.45	2,888,333	1,276	0	river hub

Community	Utility	\$/gal	Gallons	Route from Cook Inlet (sm)	River mi past Bethel	Class
Sleetmute	Middle Kuskokwim Elec	4.49	31,312	≈1,586	≈310	far Kusko
Hooper Bay	AVEC	4.50	235,572	≈1,350 (coastal)	—	coastal
Emmonak	AVEC	4.55	351,122	≈1,290 (Yukon mouth)	—	Yukon
Upper Kalskag	AVEC	4.57	96,294	≈1,344	≈68	Kusko river
Togiak	AVEC	4.61	181,852	≈1,300 (coastal)	—	coastal
Aniak	Aniak Light & Power	4.64	187,718	≈1,369	≈93	Kusko river
McGrath	McGrath Light & Power	4.96	151,194	≈1,706	≈430	far Kusko
Kwethluk	Kwethluk Inc	5.31	166,280	≈1,288	≈12	Kusko river
~~Akiak~~	Akiak City Council	~~7.95~~	96,989	—	≈25	excluded — "powerhouse down x 2 mths," 10-month report
~~Tuluksak~~	Tuluksak Traditional	~~11.85~~	56,913	—	≈35	excluded — "Diesel generation and usage x 10" reporting error

(Lower Kalskag and Mt. Village report \$0.00 / no fuel — they buy power over interties, so no purchase price exists; excluded. Akiak and Tuluksak carry explicit anomaly comments and are excluded from all fits.)

Finding 1 — the ocean-linehaul distance gradient is undetectable (in fact negative) in revealed PCE prices. Across the four clean ocean hubs, price *falls* as distance from Cook Inlet *rises*: Naknek (nearest, 1,225 sm) \$3.86 → Kotzebue (farthest, ~1,830 sm) \$3.33. A naive OLS of hub \$/gal on route sm gives a slope of $\approx -7 \times 10^{-4}$ \$/gal-mi (wrong sign; R^2 near zero). This is not a freight rate — it is dominated by carrier/volume/competition effects (Crowley vs Delta Western vs Vitus service areas, hub throughput, and how much non-diesel generation each hub blends in — Kotzebue and Nome both run large wind fleets, shrinking and cheapening their diesel buy). **The PCE cross-section therefore confirms, on margin-free data, the chapter's standing conclusion: you cannot back a positive ocean linehaul \$/gal-mi out of revealed rural prices — the linehaul signal (~\$0.3/gal) is swamped by the ~\$3–5/gal commodity+basis floor.** The engineering methods A–C remain the only sound basis for the linehaul edge.

Finding 2 — Bethel's margin-free fuel is \$4.45/gal, not \$6.72. The prior Method-D "Bethel \$6.72" was a *retail pump/heating-oil* price carrying dealer margin, storage carry, and sales tax. The utility bulk price is **\$4.45**. Against an Anchorage-area utility diesel bulk basis of roughly \$3.2–3.5/gal in the same FY2025 window (Railbelt utility diesel / OPIS rack + local delivery), the **true margin-free Bethel–Anchorage delivered-fuel differential is $\approx$ \$1.0–1.3/gal**, not the \$2.8–3.2 the retail figures implied. Distributing that ~\$1.1/gal over the 1,276-sm route gives an *upper* bound of $\approx 9 \times 10^{-4}$ \$/gal-mi all-in to Bethel (linehaul + hub + inventory), still well above the engineering linehaul because it embeds the hub tank-farm carry — but it is now ~2.5× tighter than the retail-based bound and fully consistent with Methods A–C's ~\$0.3/gal freight sitting inside a ~\$1/gal total hub adder.

Finding 3 — the river-distribution gradient is real but flatter than the invoice rate, because AVEC consolidates. Regressing the Bethel-origin Kuskokwim set (Bethel 0 mi = \$4.45 baseline; Akiachak, Kwethluk, Upper Kalskag, Aniak, Sleetmute, McGrath) on river miles past Bethel:

Village	River mi	\$/gal	Δ vs Bethel	implied \$/gal-mi
Akiachak	18	4.43	−0.02	≈0
Kwethluk	12	5.31	+0.86	(small-utility fixed cost — own genset, tiny volume)
Upper Kalskag	68	4.57	+0.12	1.8×10^{-3}
Aniak	93	4.64	+0.19	2.0×10^{-3}
Sleetmute	310	4.49	+0.04	1.3×10^{-4}
McGrath	430	4.96	+0.51	1.2×10^{-3}

Excluding the two self-supplied small utilities (Kwethluk, Akiachak — each buys its own genset fuel on a non-AVEC basis, so per-stop fixed cost dominates), the AVEC-served river villages (Upper Kalskag, Aniak, McGrath, plus AVEC's own Sleetmute-adjacent set) reveal an *incremental* river differential of only $\approx 1.0\text{--}1.5 \times 10^{-3}$ \$/gal-mi — roughly **an order of magnitude below** the chapter's ADEC/MAFA invoice rate of 1.1×10^{-2} \$/gal-mi. **This does not overturn the invoice rate; it measures a different thing.** The ADEC \$0.007/gal-map-mile is the *marginal freight cost* of moving a gallon up-river on a Crowley invoice. The PCE differential is the *net-of-everything realized price* a consolidated utility (AVEC operates one blended fuel-procurement pool across ~50 villages) actually pays — AVEC socializes river-distribution cost across its network, so an individual upriver AVEC village's *price* barely rises with distance even though its *freight cost* does. The routing model should keep the **invoice-based 1.1×10^{-2} on physical river edges** (it is the true marginal transport cost) and treat the flat PCE prices as evidence that *utility procurement consolidation*, not physics, sets rural retail — reinforcing "never calibrate transport edges to revealed prices."

Finding 4 (surprising). Sleetmute — the chapter's canonical far-upriver anchor, ~310 river miles past Bethel — pays **\$4.49/gal for utility bulk diesel, essentially identical to Bethel's \$4.45**, despite its retail heating-oil price being ~\$9.43 (the number the old Method D used). The entire ~\$5/gal gap between Sleetmute's utility fuel and its retail fuel is dealer margin, tiny-volume storage carry, and once-a-year barge working capital — **zero of it is incremental river freight** in the utility's realized cost. This is the single cleanest confirmation in the whole derivation that Western Alaska price premia are overwhelmingly non-freight.

5. Source table

Source	URL / location	Year	Provided	Status
ISER, <i>Components of Alaska Fuel Costs</i> (running head: <i>Analysis of Rural Alaska Fuel Markets</i>)	local: components of alaska fuel costs.pdf	Feb 2010	\$2.5M fixed + \$3.5M fuel/misc; \$0.19/gal; \$0.19–0.22 current → >\$0.25; rate history <\$0.09 / \$0.13–0.15; small-barge \$0.40–0.80 (\$0.60 typ.); Table 4 \$0.98 total distribution; Bethel 12-ft draft/lightering; terminal fees; hub list	✓ fetched & read (pp. 12–25)
NOAA, <i>Distances Between United States Ports</i> , 14th ed.	nauticalcharts.noaa.gov/publications/docs/distances.pdf	2025	Tables 33+35: Anchorage→Unimak Pass 688 nm; Unimak Pass→Bethel 421 nm; all hub legs (\$3)	✓ fetched; supersedes reconstruction
DCCED, <i>Current Community Conditions: Fuel Prices Across Alaska</i>	commerce.alaska.gov/.../CommunityFuelReport2005.pdf	Dec 2005	83% barge-served (Table 4); Nov-2005 retail: Bethel \$3.37 HF/\$3.61 gas; Sleetmute \$4.65/\$5.25; regional means	✓ fetched — contains no freight rates (reattributed)
ADEC / Northern Economics, <i>Cost Assessment for Diesel Fuel Transition...</i>	dec.alaska.gov/media/8915/ulsd-cost-rpt-1207.pdf	Dec 2007	River barge \$0.007/gal-map-mi (p. 21); barge sizes 63k–5.9M gal (Table 7); volume discounts (Table 8); FY06 PCE prices (Table 10); air 1.0–1.4¢/gal-mi (p. 73)	✓ fetched — no ocean-linehaul rate (reattributed to ISER)
OPPM heating-oil contract 2022-0200-4928	oppm.doa.alaska.gov/media/1337/08-heating-oil.pdf	6/12/2026 sheet	Bethel (Vitus) \$7.4970 – \$0.2800 = \$7.2170; Anchorage = OPIS LA rack + \$0.2872; Dutch Harbor \$3.55; Kodiak \$2.76; Aniak \$6.222; St. Mary's \$9.3931	✓ fetched (live sheet; earlier "\$3.7820" was a stale snapshot)
DCRA Alaska Fuel Price Report, Winter 2026	commerce.alaska.gov/.../Alaska%20Fuel%20Price%20Report%20-%20January%202026.pdf	Feb 17, 2026	Statewide gas avg \$6.63 (Summer 2025: \$6.75); Western region gas \$7.97 / HF \$7.85; gap vs national \$3.93; lock-in mechanism; ANS \$64.84/bbl	✓ fetched — summary-only; per-community cells in online DB
ISER summary / RS_68 <i>Dollars of Difference</i>	iseralaska.org (RS_68.pdf)	2008–10	Bethel \$4.58 vs Anchorage \$4.25 snapshot; "barged 1,800 mi" (rejected)	snippet only
KYUK: lock-in / first barge / Sleetmute series	kyuk.org (2026-05/06)	2026	Bethel \$6.72 (pre-shock) / \$9.37 (June 2026); Sleetmute \$9.43→\$11.89; Crowley +\$2.50/gal	secondary, retained
Alaska Beacon village-prices piece	alaskabeacon.com 2026-06-03	2026	Mountain Village \$8.71, Togiak \$7.56 (winter 2026 state survey)	secondary, retained
AAA via Stacker/Alaska's News Source	alaskasnewsresource.com 2026-05-13	2025–26	Anchorage \$3.56 (6/2025); \$5.24 (5/2026)	secondary, retained
City of Bethel – Port; Alaska Logistics; AMHS Unalaska; Professional Mariner; ADN Vitus 2012	various	—	River mileage 70–80; "–900 mi SW of Anchorage"; tug burn 3–5k gal/day; market structure	secondary, retained
AEA PCE FY2025 Statistical Report by Community (Final)	local: 2026.03.01 FY2025 PCE Statistical Report by Community (Final).pdf	3/1/2026 (FY2025 = Jul 2024–Jun 2025)	Per-community margin-free utility "Average Price of Fuel": Bethel \$4.45, Kotzebue \$3.33, Dillingham \$3.54, Nome \$3.59, Naknek \$3.86, St. Mary's \$4.06, Emmonak \$4.55, Hooper Bay \$4.50, Togiak \$4.61, Aniak \$4.64, Upper Kalskag \$4.57, Akiachak \$4.43, Kwethluk \$5.31, Sleetmute \$4.49, McGrath \$4.96 (Akiak/Tuluksak anomalous, excluded); Method E	✓ fetched & read (community pages 15–177); supersedes retail Method D
State marine diesel & unleaded contract 2022-0200-4927	local: 08-marine-diesel-fuels.pdf	sheet 6/30/2026	OPIS-rack-plus-mgmt-fee structure; coastal terminals only (no Bethel/Kuskokwim/Yukon rows); mgmt fee King Salmon \$2.55, Kodiak Dockside \$0.696, Dutch Harbor \$0.634, Homer Dockside \$0.924, Seward Dockside \$0.437 (FY27) — fee reflects volume/competition, not distance; no clean same-contract Bethel differential available	✓ fetched & read (both pp.); limited Method-D use

6. Cross-method comparison and reconciliation

Method	Linehaul \$/gal (2026\$, Cook Inlet→Bethel)	\$/gal-mile (over 1,276 sm)
A. ISER 2010 build-up (✓, CPI-adjusted)	\$0.29–0.33 market rate; \$0.50 full-cost ceiling	2.3–2.6×10 ⁻⁴ (ceiling 3.9×10 ⁻⁴)
B. Published rate history 2005/2010 (✓, CPI-adjusted)	\$0.22–0.38	1.7–3.0×10 ⁻⁴
C. Independent 2026 voyage build-up (NOAA distances)	\$0.23–0.32	1.9–2.6×10 ⁻⁴
D. Revealed retail differentials	≤ \$0.50 (bound); 2008 snapshot ≈\$0.33	≤ 3.9×10 ⁻⁴
E. PCE FY2025 margin-free utility prices (✓ NEW)	linehaul not extractable (hub gradient negative); Bethel all-in adder ≈\$1.0–1.3/gal (upper bound, embeds hub carry)	linehaul indeterminate; all-in-to-Bethel bound ≤ 9×10 ⁻⁴

Resolution of the ISER season-cost ambiguity (✓ resolved from the primary text, pp. 16–17). The exact passage: *"Current fuel distributors estimate the fixed cost of a tug and barge set at \$2.5 million for the season"* (p. 16), and *"In addition to these fixed costs, the costs of tug fuel and other miscellaneous items for a six-to-seven trip season between Cook Inlet and the Bering Sea ports is another \$3.5 million. A typical linehaul barge carries three million gallons of fuel, for a total of eighteen million gallons for a six-trip season. This results in approximately \$0.19 per gallon in cost for*

transportation only. ... The applicable math is \$3.5 million divided by 120 days, for a daily rate of almost \$30,000" (p. 17). So the text unambiguously describes **\$2.5M fixed PLUS another \$3.5M variable = \$6.0M total** ($\rightarrow$ \$0.33/gal at 18M gal), yet the report's own per-gallon and per-day arithmetic uses **only the \$3.5M** ($\rightarrow$ \$0.194/gal), and the same page states the observed market rate as **\$0.19–\$0.22/gal**, expected to rise **over \$0.25** as double-hull replacement forces re-capitalization. Reading: **\$0.19–0.22 is the verified charged linehaul rate; \$0.33 is the implied full economic cost**, and the gap is exactly the below-cost pricing the report documents elsewhere (fleet age 40 yr vs 27-yr useful life with no reinvestment, pp. 13–14; K-Sea losing half its market capitalization, p. 17). Since OPA-90 double-hull replacement completed in 2015, 2026 real rates plausibly sit **between** the escalated market rate (\$0.29–0.33) and the escalated full cost (\$0.50) — the point estimate uses the market reading; the range's upper half absorbs the full-cost reading.

Other reconciliation notes: (i) three genuinely independent methods (A, B, C) converge on **\$0.22–0.38/gal $\Rightarrow$ $\sim 1.7\text{--}3.0 \times 10^{-4}$ \$/gal-mi** over the NOAA-verified route, with the revealed-price methods (D, E) bracketing from above; (ii) retail/contract differentials (Bethel–Anchorage $\approx$ \$2.8 net of tax; Bethel–Dutch Harbor \$3.67) and now the *margin-free PCE cross-section* confirm that $\sim 75\text{--}90\%$ of Western Alaska price premia are storage/working-capital/margin/timing/procurement-consolidation, **not freight** — never calibrate transport edges to revealed prices; (iii) the 2026 Iran-war shock moves commodity price, not freight ($\sim$ \$0.02/gal pass-through), so these per-mile rates are pre/post-shock invariant in real terms; (iv) **Method E (PCE FY2025, margin-free) upgrades the revealed-price channel decisively**: it shows the ocean-linehaul distance gradient is undetectable/negative even on margin-free data (so A–C remain the only sound linehaul basis), tightens the Bethel all-in adder from the retail-based $\approx$ \$2.8 to a margin-free $\approx$ \$1.0–1.3/gal, and shows the *realized* river-price gradient ($\sim 1 \times 10^{-3}$ \$/gal-mi across consolidated AVEC villages) runs $\sim 10\times$ below the invoice-based marginal freight rate because AVEC socializes distribution — a procurement artifact, not a physics correction, so the physical river edge keeps the 1.1×10^{-2} invoice rate.

7. Final estimates (2026 USD)

(a) Ocean linehaul only — large tank-barge (2.5–3.5M gal $\checkmark$ ISER p. 15) tug-and-barge linehaul, Cook Inlet (Nikiski/Anchorage) $\rightarrow$ Western Alaska hub, per delivered gallon per statute mile of route:

- **Point:** 2.3×10^{-4} \$/gal-mi (\$0.00023; $\approx$ \$0.29/gal over the NOAA-verified 1,276-mi Anchorage $\rightarrow$ Bethel route)
- **Range:** 1.7×10^{-4} – 3.9×10^{-4} (lower = escalated 2005 rates; upper = ISER full-economic-cost reading)
- **Confidence:** medium-high $\rightarrow$ **high on the \$/gal figure** (three-method convergence, now on verified primary numbers; residual uncertainty is the CPI-vs-marine-cost escalator, not the source data)
- **Includes:** tug+barge seasonal capital/crew/insurance/maintenance, tug fuel, carrier overhead & margin, bar-crossing lightering into Bethel (embedded in the quoted hub rates). **Excludes:** hub tank-farm storage/throughput, inventory working capital, retail margin, taxes, onward village distribution.

(b) All-in delivered including river lightering / small-barge distribution:

- **To Bethel itself (hub endpoint):** $\approx 3.1 \times 10^{-4}$ \$/gal-mi ($\approx$ \$0.40/gal incl. hub offload/wharfage \$0.05–0.23 — ISER terminal fees p. 18 $\checkmark$, escalated; range $2.3\text{--}4.5 \times 10^{-4}$). (PCE FY2025 cross-check, Method E $\checkmark$: margin-free Bethel utility fuel \$4.45 vs $\sim$ \$3.2–3.5 Anchorage-basis = $\approx$ \$1.0–1.3/gal total delivered-fuel adder, of which freight is only the \$0.3–0.4 modeled here and the balance is hub tank-farm carry + inventory — consistent, and the freight portion is unmoved.)
- **To a canonical Kuskokwim village via Bethel transshipment** (~ 75 river mi beyond Bethel; total path $\approx 1,350$ mi): linehaul \$0.29 + hub transfer $\approx$ \$0.15 + river distribution 75 mi $\times$ \$0.011 $\approx$ \$0.83 $\Rightarrow \approx$ **\$1.3/gal $\Rightarrow \approx 1.0 \times 10^{-3}$ \$/gal-mi averaged over the full path** (range $0.7\text{--}1.6 \times 10^{-3}$).
- **Recommended for a per-edge routing model:** don't smear — use **2.3×10^{-4} \$/gal-mi on ocean/linehaul edges**, a **\$0.10–0.35/gal node transfer cost at the hub** (offload + dock + third-party terminal fees, ISER p. 18 $\checkmark$), and **1.1×10^{-2} \$/gal-mi on river-distribution edges** ($\checkmark$ ADEC 2007 p. 21 invoice-based \$0.007/gal-map-mi escalated; range $0.6\text{--}1.8 \times 10^{-2}$; per-stop fixed costs mean long legs run cheaper — Sleetmute's 310-mi leg reveals $\leq 0.9 \times 10^{-2}$, short legs imply $\sim 1.5 \times 10^{-2}$).
- **Confidence: medium-high** (distribution leg now rests on two independent verified anchors — ISER Tables 2–3 and the ADEC/MAFA invoice rate — that agree within 10%).

8. What would most improve this estimate

1. **$\sim$ PCE FY2024/FY2025 statistical report** (AEA) — per-community utility fuel purchase prices (bulk, margin-free) $\sim$ **$\checkmark$ DONE (Method E, 2026-07-17)**. The FY2025 report was fetched and 15 communities extracted. Result: margin-free prices cluster at \$4.4–5.3/gal for Kuskokwim/Yukon villages, ocean hubs \$3.3–3.9; the linehaul distance gradient is undetectable/negative and the river gradient ($\sim 1 \times 10^{-3}$ \$/gal-mi realized) runs $\sim 10\times$ below the invoice rate due to AVEC procurement consolidation. Confirmed the 2007 pattern (consolidation compresses village prices). Linehaul unchanged; retail-based bounds tightened.
2. **An actual current tariff or contract split into product vs freight:** Crowley/Vitus/Delta Western delivered quotes (e.g., Yupiit or LKSD bulk fuel bids, AEA Bulk Fuel Revolving Loan purchase records). The OPPM sheet gives delivered prices but no product/freight split; pairing Bethel \$7.2170 (6/12/2026) with the OPIS LA rack print for that date would yield a same-contract all-in differential.
3. **ISER 2008 Components of Delivered Fuel Prices in Alaska** (Finalfuelpricedelivered.pdf) — the June-2008 community case-study tables (only remaining unfetched study document).
4. **AIS track data** for real voyage cycles/trip counts per season (would tighten Method C's 12.7-day cycle and 165-day season).
5. A **marine towing PPI series** (BLS PPI inland/coastal water freight) to replace CPI-U as the escalator — freight-specific inflation 2010 $\rightarrow$ 2026 likely ran above CPI (post-OPA-90 capital recovery), which is why the point estimate should be read as the *lower-central* part of the stated range.

Verification addendum (2026-07-15)

Fetched and read: (1) ISER Feb-2010 report (local PDF, pp. 12–25); (2) NOAA *Distances Between United States Ports* 2025 ed. (Tables 29–36 region; Alaska Tables 33–36); (3) DCCED *Fuel Prices Across Alaska* Dec 2005 (all 16 pp.); (4) ADEC/Northern Economics ULSD cost report Dec 2007 (ToC + §§2.5–2.9, 5.1–5.2.6); (5) OPM heating-oil contract 2022-0200-4928 (all 4 pp., sheet dated 6/12/2026); (6) DCRA *Alaska Fuel Price Report: Winter 2026* (all 18 pp.). All six fetches succeeded (PDFs parsed locally).

Confirmed as originally stated: ISER \$2.5M/season fixed cost (p. 16); \$0.19/gal transportation-only and \$0.19–0.22 → >\$0.25 linehaul rates (p. 17); 3M gal × 6 trips = 18M gal; five-hub list; Bethel 12-ft controlling depth with one-third lightering (pp. 17–18); small-barge ~\$0.60/gal typical (Table 2, p. 20); DCCED 83% barge-served (2005, Table 4); DCRA Winter 2026 Western-region price levels and summer lock-in mechanism; commodity-shock treatment.

Corrected (old → new): - Route: Nikiski→Bethel 935 nm / "1,100 sm canonical" → **Anchorage→Bethel 1,109 nm = 1,276 sm** (NOAA Tables 33+35); **canonical 1,250 sm**; hub-rotation mean 1,133 sm → **1,420 sm**. - "\$0.19–0.22 (2007\$, DEC/Northern Economics)" → the figure is **ISER 2010 p. 17 (2010\$)**; DEC 2007 contains no ocean-linehaul rate. - "\$0.13–0.15 published by DCCED 2005" → **ISER 2010 p. 17 rate history (2000–2005)**; DCCED 2005 is retail-only. - Village distribution "ISER ≈\$1.00/gal (2009\$) → \$1.54 (2026\$)" → **ISER \$0.60 typ., \$0.40–0.80 (2010\$, p. 20) → \$0.91 (0.60–1.21) 2026\$**; the ~\$1 figure was Table 4's *total* distribution cost (\$0.98) including linehaul. New independent anchor: **ADEC 2007 p. 21, \$0.007/gal-map-mi river rate → 1.1×10⁻² \$/gal-mi 2026\$**. - ISER escalation basis 2009 (×1.54) → **2010 (×1.51)**. - OPM Bethel row "\$3.7820 – \$0.2800 = \$3.5020" (stale snapshot) → current sheet **\$7.4970 – \$0.2800 = \$7.2170 (6/12/2026)**; Anchorage has no flat price (OPIS LA rack + \$0.2872), so the hoped-for same-contract Bethel–Anchorage differential is not computable from the sheet alone; Dutch Harbor/Kodiak/Aniak comparators added to Method D instead. - DCRA per-community cells (Bethel \$6.72, Sleetmute \$9.43, Mountain Village \$8.71, Togiak \$7.56): **not in the PDF** (summary-only report); retained on secondary-source authority, consistent with the verified regional averages.

ISER ambiguity resolved: the passage reads "\$2.5M fixed ... another \$3.5 million" (= \$6.0M total, \$0.33/gal), but the report's own arithmetic and its observed market rate use/state \$0.19–0.22. Both readings are kept with distinct roles: **\$0.19–0.22 = charged market rate (point estimate)**; **\$0.33 = full economic cost (upper bound)** — see §6.

Effect on final estimates: per-gallon linehaul essentially unchanged (point ≈ \$0.29/gal to Bethel); per-mile linehaul point **2.6×10⁻⁴ → 2.3×10⁻⁴ \$/gal-mi** (route is 12–16% longer than reconstructed), range 1.8–4.0 → **1.7–3.9×10⁻⁴**. Bethel all-in 3.5×10⁻⁴ → **3.1×10⁻⁴**. Village-via-Bethel ≈\$2.0/gal (1.7×10⁻³) → **≈\$1.3/gal (1.0×10⁻³)** after the distribution-leg correction. River-edge rate 1.3×10⁻² (0.8–2.1) → **1.1×10⁻² (0.6–1.8)**, now invoice-anchored. Confidence upgraded on (a) and on the distribution leg.

Verification addendum — second pass (local documents, 2026-07-17)

Fetched and read: (7) AEA PCE FY2025 *Statistical Report by Community* (final, 3/1/2026; local PDF, community pages 15–177 — 15 communities extracted); (8) **State marine diesel & unleaded contract 2022-0200-4927** (local PDF, both pp., sheet 6/30/2026). Both parsed locally. This pass closes §8 item #1 (the PCE report was the top-listed remaining refinement).

New method added: Method E — the PCE "Average Price of Fuel" is the utility's total-fuel-cost ÷ gallons, i.e. the delivered bulk price **net of retail margin, sales tax, and stored-inventory working capital** — the margin-free revealed price the chapter had been missing. Dollar-year FY2025 = Jul 2024–Jun 2025 purchases, **entirely pre-shock** (the mid-2026 spike is later), so clean 2025\$.

Key extracted prices (\$/gal, FY2025): ocean hubs — Kotzebue **3.33**, Dillingham **3.54**, Nome **3.59**, Naknek **3.86**; Yukon/coastal — St. Mary's **4.06**, Hooper Bay **4.50**, Emmonak **4.55**, Togiak **4.61**; Kuskokwim river — Akiachak **4.43**, Bethel **4.45**, Upper Kalskag **4.57**, Aniak **4.64**, Sleetmute **4.49**, McGrath **4.96**, Kwethluk **5.31**. Excluded anomalies: Akiak \$7.95 ("powerhouse down x 2 mths," 10-mo report), Tuluksak \$11.85 ("Diesel generation and usage x 10" error); Lower Kalskag / Mt. Village report \$0 (intertie-fed, no fuel purchase).

Did the estimates move? - Ocean linehaul: NO CHANGE. The margin-free hub cross-section shows the distance gradient is *negative* (Naknek nearest = most expensive \$3.86; Kotzebue farthest = cheapest \$3.33) — linehaul is not extractable from revealed prices even net of margin. Engineering Methods A–C remain the sole linehaul basis; point **2.3×10⁻⁴ \$/gal-mi** and range **1.7–3.9×10⁻⁴** stand. - **River-distribution edge: NO CHANGE to the modeled rate, but its interpretation is now firmer.** The realized PCE price gradient across consolidated AVEC river villages is only ~1.0–1.5×10⁻³ \$/gal-mi — ~10× below the ADEC invoice rate — because AVEC socializes distribution cost across ~50 villages. That is a *procurement-consolidation artifact in prices*, not a correction to *marginal freight cost*. The physical river edge keeps the invoice-anchored **1.1×10⁻² \$/gal-mi**; Method E is logged as corroborating "don't calibrate edges to revealed prices," not as a competing rate. - **Bethel all-in bound: TIGHTENED (not the point estimate).** Old retail-based Method-D adder Bethel–Anchorage ≈\$2.8/gal net of tax → **margin-free ≈\$1.0–1.3/gal** (Bethel utility \$4.45 vs ~\$3.2–3.5 Anchorage basis). The all-in-to-Bethel per-mile *upper bound* tightens from ≤3.9×10⁻⁴ (Method D) to ≤9×10⁻⁴ **including hub carry**, but the **3.1×10⁻⁴ point estimate is unchanged** — freight is confirmed to be only ~\$0.3–0.4 of that ~\$1.1 adder.

Most surprising finding: Sleetmute pays \$4.49/gal for utility bulk diesel — statistically identical to Bethel's \$4.45 — despite being ~310 river miles upriver. The old Method D used Sleetmute's *retail* price (\$9.43) as a far-upriver anchor; the margin-free utility price shows **zero of the \$5/gal Sleetmute–Bethel retail gap is incremental river freight**. It is entirely dealer margin, tiny-volume storage carry, and annual barge working capital. This is the cleanest single confirmation in the derivation that Western Alaska price premia are overwhelmingly non-freight. (Runner-up: the *ocean hubs are cheaper than Bethel* — Kotzebue \$3.33 despite being ~550 sm farther from Cook Inlet — driven by hub volume and blended wind generation.)

Method D status: downgraded from primary to superseded-by-E for the Bethel differential; its structural point (retail premia » freight) is now proven on margin-free data. The marine diesel contract 2022-0200-4927 contributed little (coastal terminals only, no Bethel/river rows; its OPIS-rack-plus-mgmt-fee structure means fees track volume/competition, e.g. King Salmon \$2.55 vs Dutch Harbor \$0.63, not distance).

Plane (air fuel delivery) — \$/gallon-mile derivation

1. Search strategy

Tooling constraint (disclosure): WebFetch and Bash (curl) were both permission-blocked in the original session. The derivation below was originally built from WebSearch result snippets. **Update 2026-07-15: the primary sources have now been fetched and verified; corrections are propagated throughout and summarized in the Verification addendum at the end.** The one material error found: the Fairbanks *heating fuel* baseline was \$4.05, not \$3.34 (\$3.34 is gasoline only). **Update 2026-07-17 (second pass, local documents): the AEA FY2025 PCE Statistical Report by Community was read page-by-page; Method A" was upgraded from a four-community sketch to a mode-screened distance regression, and the Fairbanks hub anchor is now empirical (see the dated second-pass note in the Verification addendum). Point estimate held at \$0.025.**

Search sequence (all July 2026): 1. Carrier/aircraft identification: "Everts Air Fuel DC-6 bulk fuel delivery rural Alaska", C-46 tanker, "Alaska Air Fuel Wasilla DC-4". 2. Revealed prices: Alaska DCRA Fuel Price Report (Winter 2026 PDF, January 2026 survey), community-specific queries (Anaktuvuk Pass, Arctic Village, Alatna, Hughes, Lime Village, McGrath/Nikolai), AHFC fuel survey, PCE statistical reports (AEA), KYUK/ADN/Alaska Beacon/KNBA 2026 fuel-crisis coverage. 3. Operating economics: DC-6 fuel burn (PMDG/prop-liners technical data), avgas prices (AirNav July 2026), airtanker contract rates. 4. Bounding tariffs: Everts Air Cargo rates, Wright Air Service freight prices, Northern Air Cargo (NAC) rates, Alaska Air Cargo. 5. Revealed federal procurement: USAspending.gov contracts to Everts Air Fuel, Inc. 6. Distances: airmilescalculator.com (FAI–AKP, FAI–ARC), NPS/pilot sources for Coal Creek airstrip; remaining distances computed by spherical approximation from community coordinates (stated where used). 7. Benchmark study: ISER (UAA) *Components of Alaska Fuel Costs* (2010) — found the key published \$/gallon/100-air-miles figure.

2. How bulk fuel moves by air in Alaska (carriers, aircraft, pricing structure)

- **Everts Air Fuel** (Fairbanks International west ramp, plus Kenai) is the dominant carrier — "the primary source of fuel for most rural villages in the state" (alaska.org, "[Pipeline in the Sky](#)"). Aircraft: **Douglas DC-6A/B** tankers (FAA-certified internal aluminum tanks, split-load capable, 2,000–5,000 gal per Everts; 5,000–6,000 gal cited for short hauls; ISER cites "4,900-gallon DC-6") and **Curtiss C-46** tankers (roughly half a DC-6 load) for shorter/rougher strips (evertsair.com/fuel, [AOPA 2005](#)). Weight, not tank volume, binds for diesel: Everts' stated average DC-6 payload is 28,000 lb ≈ **4,000 gal of diesel** (7.05 lb/gal) or ~4,700 gal of gasoline (6.1 lb/gal).
- **Alaska Air Fuel, Inc.** (Wasilla) flies **Douglas DC-4/C-54** tankers, delivering Jet A, 100LL, mogas and heating oil statewide since 2012, often hauling Crowley-supplied product (e.g., the 2024 fatal C-54 accident was a fuel run Fairbanks→Kobuk) ([Wikipedia](#), alaskaairfuel.com).
- Very small communities/lodges get drummed or small-tank fuel in Caravan/CASA-class aircraft (e.g., Glenn Air) at much higher unit cost.
- **Pricing structure:** bulk fuel air service is sold **on a charter basis only** (Everts states this explicitly) — the customer (village corporation, city, utility, North Slope Borough, federal agency) pays for the round trip; the aircraft returns empty. Villages then retail the fuel with local tank-farm and margin costs on top. So there are two distinct "costs per gallon-mile": (a) the **charter transport component** and (b) the **delivered retail premium** (charter + local storage/handling/margin). Both are derived and kept separate.
- The DC-6/C-46/DC-4 fleet burns **100LL avgas**, a major and volatile cost input (national average \$7.44/gal, Alaska region FBO average \$10.97, July 2026, [AirNav](#)).
- Air delivery is the year-round backstop: barge-served villages that miss the barge window also fly fuel in "at even higher prices" ([KYUK, Apr 2026](#)).

3. Derivation log (steps and decisions)

1. **Identified carriers/aircraft** (Everts DC-6/C-46; Alaska Air Fuel DC-4; charter-only pricing). Decision: model the DC-6 as the reference vehicle; treat C-46/DC-4 (smaller strips) as a cost-up variant.
2. **Chose the January 2026 DCRA survey as the price baseline** (calls placed Jan 6–28, 2026) because it *predates* the June 2026 Iran/Hormuz price shock — premiums computed within one survey date net out the commodity price level entirely. Rejected June–July 2026 news prices (Hooper Bay \$8.44 gas, etc.) for the premium method because the shock was propagating unevenly through communities (and Hooper Bay is barge-served anyway).
3. **Collected air-only community prices (Jan 2026; verified 2026-07-15 against the DCRA Winter 2026 PDF and the DCRA CDO ArcGIS layers, survey calls Jan 6–28, 2026):** Anaktuvuk Pass commercial heating fuel **\$9.97** and gasoline **\$9.97** (residential HF \$1.50 is NSB-subsidized — rejected as non-market; DCRA: the subsidy "is not extended to commercial businesses"); Arctic Village heating fuel **\$15.00**, gasoline **\$10.00**; Hughes heating fuel **\$13.00**, gasoline **\$11.50**; Alatna heating fuel **\$8.50**, gasoline **\$11.50** (vendor: City of Allakaket). Hub baseline **corrected:** Fairbanks gasoline **\$3.34** (University Chevron) but heating fuel **\$4.05** (Alaska Petroleum) — the snippet-era assumption that both were \$3.34 was wrong; all HF premiums now use \$4.05. Checked and absent from the Winter 2026 survey layers: Lime Village, Takotna, Stony River, Chalkyitsik, Venetie, Birch Creek, Allakaket, Fort Yukon, Beaver, Rampart, Koyukuk, Nikolai, Bettles (no records — not surveyed or no reporting vendor). Hughes/Alatna mixed-mode ambiguity noted (occasionally reached by small Koyukuk barges in high water, but predominantly flown; retained with caveat — Alatna's HF at \$8.50 vs gasoline \$11.50, and Hughes's FY2025 PCE fuel at \$4.94, both hint at some barged product). Rejected McGrath as "air-only" — Crowley still barges 250–300k gal/yr upriver ([akbizmag](#)).
4. **Established distances:** FAI–Anaktuvuk Pass **252 mi**, FAI–Arctic Village **237 mi** (airmilescalculator.com); Coal Creek strip **124 NM = 143 statute mi** east of FAI (pilot guide). Computed FAI–Allakaket/Alatna ≈ **180 mi** and FAI–Hughes ≈ **205 mi** by spherical approximation from coordinates (assumption, ±5%).
5. **Found the ISER published benchmark (verified 2026-07-15 against the local PDF, p. 13):** "With current aviation fuel prices, the most competitive air deliveries have a transportation component of **~\$1.25 per gallon per hundred air miles**. The price is even higher for the many villages with airports unable to accommodate the 4,900 gallon DC-6 transport planes operated by Everts Air Cargo." Same page: flying fuel is only

cost-effective "to communities within a few hundred miles of Kenai or Fairbanks and then only in quantities of less than five thousand gallons." (ISER/UAA, *Analysis of Rural Alaska Fuel Markets*, Feb 17, 2010 — the document served at the [componentsoffuel3.pdf](#) URL; note it attributes the DC-6s to "Everts Air Cargo".) This is the only directly published \$/gal-mile figure found; adopted as an independent anchor (Method C-ii).

6. **Found a revealed federal charter transaction (verified 2026-07-15 via the USAspending API):** contract 140P9725F0047 — **\$11,792.50 obligated** to Everts Air Fuel, Inc. (DOI/NPS), description "FUEL AMOUNT AND TYPE - 1400 GALLONS OF UNLEADED FUEL DELIVERY LOCATION - COAL CREEK CAMP AIRSTRIP", period of performance starting Sept 18, 2025 (recorded PoP end June 30, 2030; place of performance listed as Fairbanks, AK). Adopted as Method A' (small-lot revealed price). The earlier "~\$12K rounded" band is retired — the amount is exact.
7. **Built the DC-6 cost model (Method B):** fuel burn from type data (2,000 lb/hr at long-range cruise $\Rightarrow$ ~333 gal/hr; block average assumption 400 gal/hr), avgas \$7/gal bulk (below the \$10.97 Alaska FBO retail average — Everts self-fuels in Fairbanks at volume; assumption), crew/maintenance/insurance build-up (assumptions stated in §4), 25–30% overhead+margin to convert cost to charter-equivalent. Rejected using DC-10 airtanker rates (\$26,500/hr) as irrelevant scale; no published DC-6 charter rate found.
8. **Tariff bounding (Method C-i, upgraded 2026-07-15):** NAC's published backhaul formula (**\$0.35/lb** base) gives a general-cargo floor; Everts Air Cargo's per-pound village rate table was fetched directly ([evertsair.com/cargo/rates](#)): FAI–Allakaket \$0.80/lb, FAI–Arctic Village \$0.83/lb, FAI–Anaktuvuk Pass \$0.92/lb, ANC–Bethel \$1.01–1.07/lb by weight break; fuel surcharge **26%** effective July 15, 2026) and now provides a real packaged-freight upper bracket. Later the same day, Wright Air Service's full Fairbanks-hub table and NAC's rate page were also fetched (see step 9).
9. **Second-carrier tariff cross-check (fetched 2026-07-15):** Wright Air Service ([wrightairservice.com/freight-prices/](#), 11+ lb tier): FAI–Allakaket **\$0.84/lb**, FAI–Anaktuvuk Pass **\$0.92/lb**, FAI–Arctic Village **\$0.85/lb**, FAI–Hughes **\$0.85/lb**, FAI–Fort Yukon \$0.65/lb — an independent carrier pricing the same lanes within 0–6% of Everts, which is strong evidence the per-lb tariff level is a competitive market price, not one carrier's quirk. NAC ([nac.aero/rates](#), effective 2026-01-02): backhaul formula **\$0.35/lb $\times$ fuel surcharge $\times$ 6.25% federal tax** confirmed verbatim (\$30 min); headhaul ANC–Bethel \$1.25 (1 lb) $\rightarrow$ **\$1.19/lb (5,000 lb)**, bracketing Everts' \$1.01–1.07. Alaska Air Cargo was considered and dropped — it does not carry bulk fuel (hazmat), so its tariff is not a meaningful bound.
10. **Inflation factors** (BLS CPI-U annual averages; 2025–26 assumed +2.5%/yr): 2010 $\rightarrow$ 2026 $\times$ **1.51** (218.1 $\rightarrow$ ~330); 2025 $\rightarrow$ 2026 $\times$ **1.02**. Fuel-specific escalation handled separately inside Method B via the avgas price input.
11. **Distance dependence:** derived structurally from the Method B two-part model (fixed ground/turn time vs. airborne time) and checked against the empirical spread (143-mi Coal Creek vs 252-mi AKP).
12. **Reconciled** (§7) and issued a two-part recommended edge cost plus a single-constant fallback (§8).

4. Method A / B / C — full derivations with arithmetic

Method A — Revealed delivered-price premiums (DCRA, Jan 2026; commodity-neutral; verified 2026-07-15)

Premium = village price – Fairbanks same-product price (**HF baseline \$4.05, gasoline baseline \$3.34** — verified from the DCRA CDO ArcGIS survey layers), same survey window, then $\div$ air miles:

Community	Product	Price	Premium	Miles (FAI)	\$/gal-mi (retail-inclusive)
Anaktuvuk Pass	HF (commercial)	\$9.97	\$5.92	252	\$0.0235
Anaktuvuk Pass	Gasoline	\$9.97	\$6.63	252	\$0.0263
Arctic Village	HF	\$15.00	\$10.95	237	\$0.0462
Arctic Village	Gasoline	\$10.00	\$6.66	237	\$0.0281
Hughes	HF	\$13.00	\$8.95	205	\$0.0437
Hughes	Gasoline	\$11.50	\$8.16	205	\$0.0398
Alatna	Gasoline	\$11.50	\$8.16	180	\$0.0453
Alatna	HF	\$8.50	\$4.45	180	\$0.0247

(New rows added 2026-07-15: AKP gasoline, Arctic Village gasoline, Alatna HF — all from the CDO "Gas Prices, All Years" / "Heating Fuel Price, All Years" layers, Winter 2026 records. Distances as in §3: AKP 252 mi and Arctic Village 237 mi per [airmilescalculator](#); Hughes ~205 mi and Alatna ~180 mi by spherical approximation, $\pm$ 5%. Note the within-community spreads — Arctic Village HF \$15.00 vs gasoline \$10.00, Alatna HF \$8.50 vs gasoline \$11.50 — retail idiosyncrasy and possible mixed delivery mode/stock vintage; treat single-village points as noisy.)

These include village tank-farm and retail margin. Correcting to a transport-only figure by subtracting an assumed **\$1.50/gal** local storage+distribution+margin component (ISER puts storage/distribution near \$1.00/gal typical, higher in micro-markets; assumption):

- AKP: HF (5.92–1.50)/252 = **\$0.0175**; gasoline (6.63–1.50)/252 = **\$0.0204** — the efficient case (NSB-organized full DC-6 loads)
- Arctic Village: HF 9.45/237 = **\$0.0399**; gasoline 5.16/237 = **\$0.0218**
- Hughes: HF 7.45/205 = **\$0.0363**; gasoline 6.66/205 = **\$0.0325**
- Alatna: gasoline 6.66/180 = **\$0.0370**; HF 2.95/180 = **\$0.0164** (suspect barged product — treat as low outlier)

Method A result (corrected): transport component $\approx$ \$0.016–0.040/gal-mi (2026\$), efficient full-load (AKP) $\approx$ \$0.018–0.020, small-village $\approx$ \$0.032–0.040.

Method A" — PCE utility delivered-fuel regression (upgraded 2026-07-17; supersedes the four-community sketch). The AEA *FY2025 PCE Statistical Report by Community* (published 2026-03-01; reporting period 07/01/24–06/30/25) prints, for every participating utility, an "**Average Price of Fuel**" (\$/gal) computed as annual Fuel Cost $\div$ Fuel Used (Gallons). This is the utility's *delivered purchase price*, margin-free at the retail level and inclusive of transport — exactly the quantity §9 item 4 called for. The 2026-07-15 pass pulled only four communities from the CDO layer; here the primary PDF was read page-by-page for **every plausibly air-only community** and each was mode-screened.

Hub wholesale anchor (FY2025, empirically grounded — replaces the old \$3.2–3.6 assumption). Two **road-connected** PCE communities appear in the same report and receive trucked diesel from the Fairbanks rack over ~130–160 road miles: **Circle \$3.58** (32,675 gal) and **Central \$3.66** (36,391 gal). Their mean **\$3.62/gal** is the closest in-dataset proxy for FY2025 Fairbanks-rack-plus-short-road diesel. Adopt **\$3.60/gal** as the Interior (Fairbanks-hub) anchor, uncertainty ±\$0.30; it carries a small road-haul component, so it biases the fixed intercept *slightly low* (conservative). No comparable road anchor exists for the Kotzebue hub (Kotzebue is itself a high-cost coastal barge point), so NW-Arctic points are treated as a **qualitative** cross-check only, not entered into the regression.

Air-only community table (FY2025 PCE "Average Price of Fuel"; premium = price – hub anchor; distances great-circle from community/airport coordinates, spherical approximation ±5%, except AKP/Arctic Village which use airmilescalculator):

Community	Utility	FY25 \$/gal	Gallons	Hub	Air mi	Premium \$/gal	\$/gal-mi
Anaktuvuk Pass	North Slope Borough	\$6.38	320,380	FAI (\$3.60)	252	2.78	\$0.0110
Allakaket; Alatna	Alaska Power & Telephone	\$6.19	56,927	FAI	185	2.59	\$0.0140
Nikolai	City of Nikolai	\$7.65	37,854	FAI	233	4.05	\$0.0174
Takotna	Takotna Comm. Assoc.	\$5.63	34,801	FAI	285	2.03	\$0.0071 △
Hughes	Hughes Power & Light	\$4.94	53,418	FAI	202	1.34	\$0.0066 △
Ambler	AVEC	\$6.17	88,715	OTZ	126	—	— ◇
Shungnak (+Kobuk intertie)	AVEC	\$7.77	106,514	OTZ	138	—	— ◇
Noatak	AVEC	\$9.41	134,327	OTZ	55	—	— ◇

△ **Excluded from regression — mode-contaminated.** Takotna sits ~18 mi from barge-served McGrath and its \$5.63/285 mi *falls* below shorter-haul villages (breaks the distance monotonicity) → almost certainly receives McGrath-staged barged product. Hughes (\$4.94, down from FY2024 \$6.41) is on the Koyukuk and its price collapsed to barge-level → partial barge in FY2025 (chapter already flagged this). ◇ **NW-Arctic (Kotzebue hub) — flagged ambiguous, cross-check only:** Ambler and Shungnak are upper-Kobuk river-barge villages from Kotzebue in normal water, flown only in low-water years; the FY2025 average blends whatever mode ran. Noatak (\$9.41, 55 air mi from OTZ) is the striking datum — an *extreme* \$/gal-mi at short haul, exactly the fixed-cost-dominated short-haul behavior Method B predicts, but the missing Kotzebue anchor bars a clean number.

Excluded outright (no usable FY2025 fuel datum): Arctic Village (Avg Price of Fuel \$0.00 — "11 rpts filed, no powerhouse/usage rpts," staffing), Lime Village (\$0.00, no powerhouse reports), Venetie (\$0.00, "no diesel gen/usage reprt"), Kobuk (\$0.00 — "Receives power from Shungnak via intertie"). **Confirmed barge/mixed and excluded:** Stony River (\$4.49, Middle Kuskokwim — barge), Bettles/Evansville (\$4.61, Koyukuk — barge-level price), Rampart (\$6.04, Yukon River — barge), Fort Yukon (\$6.88, Yukon barge hub). These barge/road points (\$3.58–\$4.61 for road/river vs \$6.19–\$7.65 for the clean air villages) sharply validate the mode screen: **flown fuel carries a \$2–4/gal premium over river/road fuel at comparable Interior latitudes.**

Two-part regression (\$9 item 4), clean Interior air-only set (n=3: AKP, Allakaket, Nikolai): OLS of premium (y, \$/gal) on air-miles (x): $\bar{x}=223.3, \bar{y}=\$3.14; \Sigma(dx\cdot dy)=19.56, \Sigma dx^2=2384.7 \Rightarrow \text{slope} = \text{\$0.0082/gal-mi (variable), intercept} = \text{\$1.31/gal (fixed)}$. Fit is weak ($R^2 \approx 0.12$, 3 noisy single-village points, Nikolai high-leverage) — the regression is *illustrative, not decisive*; the identified structure remains Method B's. Interpretation: this set is dominated by **organized utility full-load buys** (AKP alone is 320k gal/yr), so both terms land *below* Method B's small-lot-inclusive \$0.0126 variable / \$2.18 fixed — precisely the "utility-scale airlift is cheaper than the marginal small-lot rate" result. The per-point \$/gal-mi (0.011–0.017) clusters at the **low end** of the chapter's range. Anchor sensitivity: a \$3.30 (vs \$3.90) hub anchor shifts every premium by +\$0.30 (–\$0.30), moving the intercept by ±\$0.30 and the mean \$/gal-mi by roughly ±\$0.0014 — the ±\$0.30 anchor uncertainty is the dominant error, not the fit.

Method A" result: FY2025 utility-scale full-load air transport ≈ \$0.011–0.017/gal-mi (2026\$), fixed ≈ \$1.3/gal + variable ≈ \$0.008/gal-mi from the (noisy) regression — confirming organized bulk airlift sits at or below the Method B full-load curve, and empirically anchoring the Fairbanks hub at \$3.6/gal from the report's own road-served communities.

Method A' — Revealed federal charter transaction (NPS Coal Creek, Sept 2025; verified 2026-07-15)

\$11,792.50 ÷ 1,400 gal = \$8.42/gal delivered unleaded (exact obligated amount from USAspending; supersedes the "≈\$12K → \$8.57" estimate). Fairbanks retail unleaded ≈ \$3.45 (late 2025 assumption; Jan 2026 survey \$3.34). Transport+service premium = 8.42 – 3.45 = **\$4.97/gal** over 143 mi = **\$0.0348/gal-mi** (×1.02 to 2026\$ → **\$0.0355**; band \$0.034–0.037 now driven only by the Fairbanks price assumption, the award amount being exact). This is a genuine arms-length price for a **small-lot (1,400 gal) short-haul** delivery including carrier margin — it directly evidences the fixed-cost inflation at short distance/small load.

Method B — DC-6 operating-economics build-up (2026\$)

Inputs (assumptions flagged): - Payload: 28,000 lb ⇒ **4,000 gal diesel** (7.05 lb/gal) [Everts spec; ISER's 4,900 gal applies to gasoline/short-haul] - Block speed **230 mph** (cruise ~280, minus climb/descent/pattern) [assumption] - Fuel burn **400 gal/hr** block avg (type data: 333 gal/hr LRC to ~450 normal cruise) × avgas **\$7.00/gal** bulk = **\$2,800/hr** - Oil (R-2800s): ~\$150/hr; crew (2–3): ~\$450/hr; maintenance (4 geriatric radials, overhauls ~\$60–80K/engine/~1,200 hr + airframe): ~\$1,000–1,300/hr; insurance/ownership: ~\$250/hr [assumptions] - **Direct operating cost ≈ \$4,650/hr**; +25% overhead & margin ⇒ **charter-equivalent ≈ \$5,800/hr** - Fixed per-round-trip ground/turn time (load, pump-off at ~150–200 gpm, taxi): **1.5 hr** [assumption]

Trip cost to distance D: $\$5,800 \times (2D/230 + 1.5)$. Per gallon-mile = that ÷ (4,000 × D):

- Variable term: $2 \times \$5,800 / (230 \times 4,000) = \text{\$0.0126/gal-mi}$
- Fixed term: $\$5,800 \times 1.5 / 4,000 = \text{\$2.18/gal per delivery, } \div D$

D (mi)	\$/gal	\$/gal-mi
100	\$3.44	\$0.0344
143 (Coal Creek)	\$3.98	\$0.0278
200	\$4.70	\$0.0235
252 (AKP)	\$5.36	\$0.0213
400	\$7.22	\$0.0180

Sensitivity: avgas \$6→\$8/gal moves the hourly cost $\mp/\pm \sim 9\%$ ($\approx \pm \$0.002/\text{gal-mi}$ at 250 mi). C-46/DC-4 variant (payload ~2,200 gal, \$3,500/hr, 180 mph): variable \$0.0177 + \$1.91/D fixed = **\$0.027/gal-mi at 200 mi** — matching the small-village Method A band.

Method B result: \$0.018–0.028/gal-mi full DC-6 loads over 150–400 mi; ~\$0.027–0.035 for C-46/DC-4-class or <150-mi hauls.

Method C — Published tariff/benchmark bounds

(i) General air-cargo bounds (NAC floor + Everts actuals, fetched 2026-07-15): NAC's documented backhaul base rate **\$0.35/lb**; ANC–Bethel = 399 mi. $0.35 \times 7.05 \text{ lb/gal} = \$2.47/\text{gal} \div 399 = \mathbf{\$0.0062/\text{gal-mi}}$ — a floor *not achievable* for bulk fuel (specialized tanker, guaranteed empty return, hazmat); it explains why fuel-by-air must exceed $\sim \$0.01$. Everts Air Cargo's actual published general-freight rates (evertsair.com/cargo/rates, fetched 2026-07-15; **fuel surcharge 26%** effective 2026-07-15, post-Hormuz-shock; minimum charge \$40): FAI–Allakaket \$0.80/lb, FAI–Arctic Village \$0.83/lb, FAI–Anaktuvuk Pass \$0.92/lb, FAI–Fort Yukon \$0.60/lb, FAI–Kaktovik \$2.12/lb; ANC–Bethel \$1.07 (500 lb) → \$1.01/lb (>5,000 lb). Diesel-equivalents incl. surcharge ($\times 1.26 \times 7.05 \text{ lb/gal} \div \text{miles}$): FAI–ARC **\$0.0311**, FAI–AKP **\$0.0324**, FAI–Allakaket **\$0.0395**, ANC–Bethel bulk **\$0.0225/gal-mi**. Packaged general freight thus brackets **\$0.022–0.040/gal-mi** — an upper-bound proxy for fuel (palletized freight carries handling costs a tanker avoids, but no empty-return guarantee), and it lands exactly on the small-lot cluster from Methods A/A'. Still labeled a **bounding proxy**, not a fuel rate.

(ii) ISER 2010 published benchmark (quote verified 2026-07-15 against the PDF, p. 13): "With current aviation fuel prices, the most competitive air deliveries have a transportation component of $\sim \$1.25$ per gallon per hundred air miles. The price is even higher for the many villages with airports unable to accommodate the 4,900 gallon DC-6 transport planes operated by Everts Air Cargo." (ISER/UAA, *Analysis of Rural Alaska Fuel Markets*, Feb 17, 2010.) \$0.0125/gal-mi in 2009–10\$; CPI-U 2010→2026: 330/218.1 = 1.51 ⇒ **\$0.0189/gal-mi (2026\$)**. Sits within $\sim 10\%$ of Method B's full-load figure — strong independent convergence.

5. Source table

Source	URL	Year	Provided	Reliabil
Alaska DCRA Fuel Price Report, Winter 2026	https://www.commerce.alaska.gov/web/Portals/4/pub/RA/Fuel_price_report/Alaska%20Fuel%20Price%20Report%20-%20January%202026.pdf ; https://storymaps.arcgis.com/stories/64310ec78e234eb3bc8f50ed16a0f37d	Jan 2026	AKP commercial HF \$9.97 / residential \$1.50 (NSB table); Arctic Village \$15.00 & Hughes \$13.00 (highest-HF list); Fairbanks gasoline \$3.34; survey dates Jan 6–28; NSB subsidy note	Verified 2026-07 15 (PDF fetched read; no full commur table in PDF)
DCRA CDO ArcGIS survey layers (Heating Fuel Price / Gas Prices / PCE, All Years)	https://maps.commerce.alaska.gov/server/rest/services/Services/CDO_Uilities/MapServer (layers 6, 7, 4)	Jan 2026	Fairbanks HF \$4.05; Alatna HF \$8.50; Arctic Village gas \$10.00; AKP gas \$9.97; Hughes \$13.00/\$11.50; PCE utility fuel prices FY2023–25	Verified 2026-07 15 (queried REST A directly)
AEA, FY2025 PCE Statistical Report by Community (Final)	local: cost_derivation_resources/2026.03.01 FY2025 PCE Statistical Report by Community (Final).pdf (183 pp); akenergyauthority.org/pce	Mar 2026 (FY2025)	Per-utility "Average Price of Fuel" \$/gal + gallons: AKP \$6.38/320,380; Allakaket/Alatna \$6.19; Nikolai \$7.65; Takotna \$5.63△; Hughes \$4.94△; Ambler \$6.17; Shungnak \$7.77; Noatak \$9.41; road anchors Circle \$3.58, Central \$3.66; barge Stony River \$4.49, Bettles \$4.61, Rampart \$6.04, Fort Yukon \$6.88	Verified 2026-07 17 (primary PDF rea page-by page)
ISER/UAA, Analysis of Rural Alaska Fuel Markets	https://iseralaska.org/static/legacy_publication_links/componentsoffuel3.pdf ; local copy components of alaska fuel costs.pdf	Feb 17, 2010	" $\sim \$1.25$ per gallon per hundred air miles"; 4,900-gal DC-6 (Everts Air Cargo); <5,000-gal	Verified 2026-07 15 (p. 1: exact quote)

Source	URL	Year	Provided	Reliability
(Components of Fuel Costs series)			lots, few hundred mi of Kenai/Fairbanks; storage/distribution components	
USAspending, Everts Air Fuel NPS contract	https://www.usaspending.gov/award/CONT_AWD_140P9725F0047_1443_140P9725A0010_1443 (data via api.usaspending.gov)	Sept 2025	\$11,792.50 / 1,400 gal unleaded to Coal Creek Camp airstrip; DOI/NPS; PoP start 2025-09-18	Verified 2026-07 15 (exact amount)
Everts Air — bulk fuel page & fleet	https://evertsair.com/fuel	2026	Charter-only pricing; 2,000–5,000 gal tanks; DC-6 payload 28,000 lb	High
AOPA "Legendary Aircraft"	https://www.aopa.org/news-and-media/all-news/2005/november/pilot/legendary-aircraft-extraordinary-service	2005	DC-6 5,000–6,000 gal (distance-dependent); North Slope ops	Med
alaska.org "Pipeline in the Sky"	https://www.alaska.org/detail/pipeline-in-the-sky	n.d.	Everts primacy, hubs (FAI/Kenai), FAA-certified tanks	Med
Alaska Air Fuel / C-54 crash article	https://en.wikipedia.org/wiki/2024_Alaska_Air_Fuel_Douglas_C-54_crash ; https://www.alaskaairfuel.com/	2024–26	Second carrier, DC-4s, FAI→Kobuk run	High
AirNav fuel price report	https://www.airnav.com/fuel/report.html	Jul 2026	100LL national \$7.44, Alaska region \$10.97	High (re FBO, no bulk)
DC-6 type/fuel-burn data	http://www.prop-liners.com/dc6btech.htm ; PMDG forum	—	2,000 lb/hr LRC burn	Med
Air Miles Calculator	https://www.airmilescalculator.com/distance/fai-to-akp/ ; /fai-to-arc/	2026	252 mi, 237 mi	High
Coal Creek strip location	https://sites.google.com/site/bushplaneparksalaska/home/alaska-fly-in-parks/yukon---charley-rivers	—	124 NM (143 mi) E of FAI	Med
NAC rates	https://www.nac.aero/rates/	eff. 2026-01-02	\$0.35/lb backhaul formula (verified verbatim); headhaul ANC–Bethel \$1.19–1.25/lb	High (fetched 2026-07 15)
Wright Air Service freight prices	https://wrightairservice.com/freight-prices/	2026	FAI-hub per-lb table: Allakaket \$0.84, AKP \$0.92, Arctic Village \$0.85, Hughes \$0.85 — matches Everts within 0–6%	High (fetched 2026-07 15)
Everts Air Cargo rates	https://evertsair.com/cargo/rates	Jul 2026	FAI–village per-lb rates (Allakaket \$0.80, Arctic Village \$0.83, AKP \$0.92, Fort Yukon \$0.60, Kaktovik \$2.12); ANC–Bethel \$1.01–1.07; fuel surcharge 26% (eff. 2026-07-15); \$40 minimum	Verified 2026-07 15
KYUK/ADN/Beacon 2026 fuel-crisis series	https://www.kyuk.org/economy/2026-04-20/... ; https://alaskabeacon.com/2026/06/03/...	2026	Shock context; air = backstop at higher price; 1,200 gal/person/yr	High (context)
KNBA — AKP fuel shortage	https://www.knba.org/news/2026-01-15/a-weeklong-fuel-shortage-in-anaktuvuk-pass-caused-school-closures-and-left-homes-without-heat	Jan 2026	18,500 gal diesel flown in a week; Everts diesel-capable fleet expansion	High
akbizmag "No Port No Problem"	https://www.akbizmag.com/industry/transportation/no-port-no-problem/	2021	McGrath still barge-served (excluded from air-only set)	Med

6. Distance-dependence analysis

Cost per gallon-mile is NOT constant — it falls hyperbolically with distance. Structurally (Method B): per-gal-mi = **\$0.0126 + (\$2.18/D)** for DC-6 loads. The fixed component (loading, pump-off, taxi, dispatch — ~1.5 hr of a ~\$5,800/hr asset spread over 4,000 gal) contributes \$2.18/gal *per delivery* regardless of distance. Consequences: per-gal-mi at 100 mi (\$0.034) is ~1.9× the 400-mi value (\$0.018); halving load size (C-46) raises the whole curve ~35–50%. The empirical points agree: Coal Creek (143 mi, 1,400-gal lot) ≈ \$0.035 vs AKP (252 mi, full loads) ≈ \$0.018–0.026 (verified values). **For**

the routing model, a two-part edge cost (fixed per-delivery \$ + true per-mile \$) is strictly better than any single \$/gal-mi constant. Beyond ~400–500 mi, aircraft fuel uplift begins to displace payload, so the curve flattens then rises slightly; irrelevant at typical hub–village stage lengths (100–300 mi).

7. Cross-method comparison and reconciliation

Method	Basis	\$/gal-mi (2026\$)	Includes
A (DCRA premiums, corrected – \$1.50/gal local)	market prices (verified)	0.016–0.040 (AKP 0.018–0.020)	charter + residual local costs
A" (PCE utility fuel, FY2025 report)	utility purchase prices, mode-screened; hub anchored to road-served PCE communities (\$3.60)	≈0.011–0.017 (clean full-load buys); regression \$1.31 fixed + \$0.0082/gal-mi (R²≈0.12)	charter, utility-scale
A' (NPS Coal Creek, \$11,792.50)	actual contract (verified)	0.034–0.037 @143 mi, 1,400 gal	charter + margin, small lot
B (DC-6 build-up)	engineering	0.018–0.028 (full loads, 150–400 mi)	charter-equivalent
C-i (NAC floor; Everts general-freight actuals)	tariffs (Everts verified)	floor 0.006–0.012; packaged-freight bracket 0.022–0.040	general cargo bounds (not fuel rates)
C-ii (ISER x1.51)	published study (verified)	0.019 ("most competitive")	charter, full DC-6

Convergence is good once load size and distance are controlled: **full-tanker-load charter cost clusters at \$0.018–0.022/gal-mi** (B, C-ii, A-efficient AKP; the FY2025 PCE regression puts organized bulk buys slightly *below* this, ~\$0.011–0.017, with the road-served-community hub anchor now empirical rather than assumed); **small-lot/small-aircraft reality clusters at \$0.031–0.040** (A', A small villages, Everts packaged-freight equivalents); the general-cargo floor confirms air fuel can't go below ~\$0.012. Retail-inclusive delivered premiums run \$0.024–0.046. No method is contaminated by the June 2026 commodity shock (Jan 2026 premiums are same-date differentials; Method B's avgas input is a stated sensitivity; the one post-shock input, Everts' 26% fuel surcharge, is flagged and only affects the bounding proxy).

8. Final estimate: point, range, confidence, includes/excludes

- **Point estimate: \$0.025 per gallon-mile (2026\$)** — carrier charter transport cost of bulk fuel by air, hub→village, averaged over the realistic mix of DC-6 full loads and C-46/DC-4 partial loads at typical 150–300 mi stage lengths.
- **Preferred model form:** edge cost = **\$0.0126/gal-mi + \$2.20/gal fixed per air delivery** (DC-6 class); multiply both terms by ~1.4 for communities whose strips only take C-46/DC-4-class aircraft.
- **Range: \$0.015–0.042/gal-mi** (\$0.015–0.022 efficient full-load long-haul, with PCE utility data supporting the low end; \$0.031–0.040 small-lot/short-haul/small-aircraft). **Delivered-price premium including village storage, handling and margins: \$0.024–0.046/gal-mi.**
- **The point estimate did not move** (\$0.025 before and after both passes). The 2026-07-15 verification adjustments were symmetric; the 2026-07-17 PCE regression added information at the **low end only** (utility full-load buys at \$0.011–0.017), which the blended point estimate already reflected by averaging efficient full loads against small-lot/small-aircraft realities (\$0.031–0.040). No basis to move \$0.025; if anything the new data *firm up* the full-load floor. The one substantive upgrade is qualitative: the Fairbanks hub anchor is now empirical (\$3.60 from road-served PCE communities), not assumed.
- **Confidence: medium-high** (held; the FY2025 PCE regression corroborates rather than overturns). The formerly snippet-only load-bearing values are now verified: Fairbanks baselines (\$3.34 gas / \$4.05 HF), all village prices, the NPS award (\$11,792.50 exact), the ISER quote, and Everts' rate table. Remaining softness: Method B's hourly cost is a build-up, not a quoted rate; Hughes/Alatna mode ambiguity; ±5% spherical distances for Hughes/Alatna.
- **Includes:** round-trip aircraft operation (empty return), crew, loading/pump-off, carrier overhead and margin. **Excludes:** fuel commodity price, hub terminal storage, village tank-farm and retail margin (except in the "delivered premium" variant above), weather-delay/emergency surcharges.
- Air is roughly **10–20× barge linehaul** on a per-gallon-mile basis (ISER barge ≈ \$0.19/gal over long linehauls).

9. What would most improve this estimate

(Updated 2026-07-15 — the original item 1 fetch list is done except where noted; see Verification addendum.)

1. ~~Remaining unfetched tariffs~~ **Done (2026-07-15 second pass):** Wright Air and NAC rate pages fetched; both corroborate the Everts-based bracket (see log steps 8–9). Alaska Air Cargo dropped — it does not carry bulk fuel (hazmat), so its tariff is not a meaningful bound. Still open: the Everts recipient profile on USAspending (all awards) could yield more \$/gal/distance points beyond Coal Creek.
2. A direct **charter quote from Everts Air Fuel / Alaska Air Fuel** (e.g., FAI→AKP, 4,000 gal diesel) would replace Method B's hourly build-up with a market rate.
3. North Slope Borough budget documents for the AKP annual airlift (gallons and contract \$) would give the single best full-load revealed rate (PCE shows NSB's AKP utility fuel at \$6.33–6.38/gal FY2024–25 on ~320k gal/yr — a contract figure would decompose this).
4. ~~Regressing PCE delivered fuel cost on air distance across all flown-fuel communities... A contemporaneous Fairbanks rack/wholesale diesel series is needed to anchor the premiums.~~ **Done (2026-07-17 second pass, local documents):** the *FY2025 PCE Statistical Report by Community* (AEA, 2026-03-01) was read community-by-community; every plausibly air-only village was extracted and mode-screened; the hub anchor is now **empirical** (\$3.60/gal, from road-served PCE communities Circle \$3.58 / Central \$3.66 in the same report — the old \$3.2–3.6 assumption is retired). The two-part regression ran (n=3 clean Interior air-only: fixed \$1.31/gal + variable \$0.0082/gal-mi, R²≈0.12 — noisy but confirming utility-scale full loads sit *below* the Method B curve). **Still open:** the clean set is small because most small air-only villages either don't report fuel (Arctic Village, Lime, Venetie — \$0.00) or show barge contamination (Takotna, Hughes); a Kotzebue-hub diesel anchor would unlock the NW-Arctic points (Ambler/Shungnak/Noatak); and verified great-circle distances (vs the ±5% spherical ones used here) would tighten the slope.

5. Verified great-circle distances for Hughes and Alatna (currently spherical approximations, $\pm 5\%$).

Verification addendum (2026-07-15)

WebFetch became available; all \$9-item-1 primary sources except Wright Air/NAC/Alaska Air Cargo were fetched and the numbers propagated.

Fetched: (1) DCRA Winter 2026 PDF (18 pp — narrative + NSB table only; no full community price table); (2) the DCRA Community Database Online ArcGIS REST layers behind the report (CDO_Uilities MapServer layers 6 "Gas Prices, All Years", 7 "Heating Fuel Price, All Years", 4 "PCE Program"), queried directly for Winter 2026 / FY2023–25; (3) USAspending API for award CONT_AWD_140P9725F0047_1443_140P9725A0010_1443; (4) evertsair.com/cargo/rates; (5) local ISER PDF p. 13 (*Analysis of Rural Alaska Fuel Markets*, Feb 17, 2010).

Confirmed: AKP commercial HF \$9.97 / residential \$1.50 (NSB-subsidized; subsidy "not extended to commercial businesses"); Arctic Village HF \$15.00; Hughes HF \$13.00 / gas \$11.50; Alatna gas \$11.50; Fairbanks gasoline \$3.34; survey window Jan 6–28, 2026; NPS award = Everts Air Fuel, 1,400 gal unleaded, Coal Creek Camp airstrip, PoP start 2025-09-18; ISER "~\$1.25 per gallon per hundred air miles" and 4,900-gal DC-6 (attributed to Everts Air Cargo), quoted exactly.

Corrected: Fairbanks *heating fuel* baseline \$3.34 → **\$4.05** (the \$3.34-for-both assumption was wrong, as suspected) — all HF premiums re-derived (AKP 0.0263→0.0235, Arctic Village 0.0492→0.0462, Hughes 0.0471→0.0437 retail-inclusive); NPS award \$12,000 → **\$11,792.50** (\$8.57→\$8.42/gal; A' 0.0366→0.0355); ISER source title corrected.

Added: three Method A rows (AKP gasoline \$9.97 → 0.0263; Arctic Village gasoline \$10.00 → 0.0281; Alatna HF \$8.50 → 0.0247 retail-inclusive); Method A" PCE utility fuel prices (AKP \$6.33–6.38, Alatna/Allakaket \$6.19–7.38, Hughes \$4.94–6.41, FY2024–25); real Everts general-freight bracket for Method C-i (FAI–ARC/AKP/Allakaket \$0.80–0.92/lb + 26% surcharge ⇒ \$0.031–0.040/gal-mi packaged-freight equivalents; ANC–Bethel bulk \$0.0225). Communities checked and absent from the Winter 2026 survey: Lime Village, Takotna, Stony River, Chalkyitsik, Venetie, Birch Creek, Allakaket, Fort Yukon, Beaver, Rampart, Koyukuk, Nikolai, Bettles.

Net effect: final point estimate **unchanged at \$0.025/gal-mi (2026\$)**; range tightened to \$0.015–0.042; confidence raised medium → medium-high. One fetch failure: maps.commerce.alaska.gov rejects the /arcgis/rest/ path (WAF) — the /server/rest/ path works.

Second pass — local documents (2026-07-17)

Read the primary source **cost_derivation_resources/2026.03.01 FY2025 PCE Statistical Report by Community (Final).pdf** (AEA, 183 pp) page-by-page to execute \$9 item 4 (Method A" upgraded from a four-community sketch to a mode-screened regression).

Fetched/read: per-community pages for every plausibly air-only village plus road/river/barge controls — Anaktuvuk Pass (p.21), Allakaket;Alatna (19), Arctic Village (25), Ambler (20), Kobuk (89), Shungnak (151), Noatak (120), Lime Village (100), Nikolai (118), Takotna (161), Venetie (178), Stony River (160), Bettles;Evansville (29), Rampart (140), Hughes (75), plus road/hub controls Central (32), Circle (43), Fort Yukon (62). Field used: **"Average Price of Fuel" = Fuel Cost ÷ Fuel Used (Gallons)** — the utility's delivered, margin-free purchase price.

Confirmed / new data: FY2025 utility fuel \$/gal — AKP **\$6.38** (320,380 gal, NSB), Allakaket/Alatna **\$6.19** (56,927), Nikolai **\$7.65** (37,854), Ambler \$6.17 (88,715), Shungnak \$7.77 (106,514, feeds Kobuk via intertie), Noatak \$9.41 (134,327); road-served controls Circle \$3.58, Central \$3.66; barge/river controls Stony River \$4.49, Bettles/Evansville \$4.61, Rampart \$6.04, Fort Yukon \$6.88.

Corrected / retired: the Method A" Fairbanks-wholesale anchor **\$3.2–3.6 (assumed)** → **\$3.60 (empirical)**, taken from the report's own road-served PCE communities (Circle/Central mean \$3.62). The FY2024/FY2025 CDO figures from the 2026-07-15 pass are consistent with the FY2025 report values.

Mode-screen results (new): *air-only*, *clean* = AKP, Allakaket/Alatna, Nikolai. *Excluded* — *no fuel datum* = Arctic Village, Lime Village, Venetie, Kobuk (all \$0.00 / intertie / non-reporting). *Excluded* — *barge/mixed contamination* = Takotna (\$5.63, McGrath-adjacent, breaks distance monotonicity), Hughes (\$4.94, down from \$6.41 → partial barge on the Koyukuk), Stony River/Bettles/Rampart/Fort Yukon (river/barge). *Flagged ambiguous, cross-check only* = Ambler/Shungnak/Noatak (Kotzebue hub, no clean diesel anchor; Noatak's \$9.41 at 55 air-mi is a textbook short-haul fixed-cost spike).

Regression (\$9 item 4): premium = price – \$3.60, on great-circle air-miles (spherical approximation $\pm 5\%$; AKP 252 & Arctic Village 237 from airmilescalculator). $n=3$ clean Interior air-only ⇒ **fixed \$1.31/gal + variable \$0.0082/gal-mi, $R^2 \approx 0.12$** — noisy (3 single-village points) but directionally clear: utility-scale full loads sit *below* Method B's \$2.18 fixed + \$0.0126 variable, consistent with organized bulk airlift being the efficient end of the range. Per-point \$/gal-mi 0.011–0.017. Dominant uncertainty is the $\pm$ \$0.30 hub anchor, not the fit.

Net effect (second pass): final point estimate **unchanged at \$0.025/gal-mi (2026\$)**; the new evidence lands entirely in the already-modeled full-load low end (\$0.011–0.017) and does not shift the blended point. Substantive gains are (a) an *empirical* Fairbanks hub anchor replacing an assumption, and (b) a mode-screened, distance-resolved confirmation that flown fuel carries a \$2–4/gal premium over river/road fuel at comparable Interior latitudes. Confidence held at medium-high. Limitation: clean air-only sample is small ($n=3$) because most tiny air villages don't report fuel or are barge-contaminated; a Kotzebue-hub anchor and verified distances remain the highest-value next steps.

Ice road / winter trail — \$/gallon-mile derivation

Method note: The original derivation (20 WebSearch snippets, WebFetch permission-blocked) was verified on 2026-07-15 with a manual-fetch round: the DCRA Winter 2026 PDF, the UIC AKP haul write-up, the Alaska Business CWAT feature, the Northern Policy Institute article, the GNWT resupply release, the Tibbitt–Contwoyto JV/Wikipedia pages, and the full 267-page ASTAR Atqasuk–Utqiagvik road study were read in full; the OMB FY2020 CWAT project sheet resisted direct fetch (certificate error; akleg mirror 403) but its appropriation amount was recovered from the search index of the same PDF.

Corrections from the verification round are integrated throughout and logged in §3 (step 8). **The recommended rate did not move: \$0.010/gal-mi.** Sub-band numbers shifted slightly (details in §7).

Second-pass note (local primary documents, 2026-07-17): Four downloaded primary PDFs were read in full — the 2012 Northwest Arctic Borough *Noatak Winter Fuel Haul System* legislative grant proposal (TPS Report 58673v1), the OMB FY2020 CWAT project sheet (AMD 62733, the one that had resisted fetch), the *North Slope Borough FY2026-2027 Budget Book*, and the Alaska PCE FY2025 Statistical Report. The Noatak proposal is the first Alaska primary source that lays out a single-tractor overland fuel-haul system in engineering detail (route length, payload, speed, trips/season, annual demand, capital cost) — it **corroborates every physical assumption in Method A2** (payload, speed, single-tracked-unit architecture) but, being a self-sustaining-business proposal, **states no operating \$/gal or \$/gal-mi**, so A2's ~\$900/hr convoy rate remains the one unverified load-bearing input. The NSB budget publishes no per-community gallons; the PCE report supplies them and now anchors the §7 traffic weighting that was previously a coin-flip. The OMB sheet text confirmed the \$1,672,077 amount and that CWAT is a traveler-safety/trail-maintenance program (not fuel-haul funding), sharpening the §8 adder. **Net effect on the recommended rate: unchanged at \$0.010/gal-mi** — the real gallon-weighted traffic mix (§7) lands the point estimate on essentially the same value the equal-weight assumption produced. Details logged in §3 (step 9) and the §10 addendum.

1. Search strategy

The plan was: (1) establish the operational reality of winter fuel hauls on the three named route archetypes — Deadhorse→Nuiqsut engineered ice road, Utqiagvik→Atqasuk tundra snow trail (CWAT), and the Kuskokwim River ice road out of Bethel; (2) then triangulate cost with four independent evidence families: (A) engineering cost build-up from equipment/labor/speed/payload, (B) revealed retail fuel-price differentials between winter-road-served and hub communities (Alaska DCRA Fuel Price Report, GNWT community fuel prices, Ontario First Nations winter roads), (C) circumpolar heavy-haul winter-road tonne-km economics (Tibbitt–Contwoyto), and (D) documented real-world haul events as bounds (the Feb–Mar 2026 Anaktuvuk Pass emergency overland haul). Twenty searches were run against news archives (KYUK, KNBA, ADN, Alaska Beacon), state PDFs (DCRA, Alaska OMB, DOT&PF), North Slope Borough budget documents, GNWT releases, and Canadian sources, followed by the 2026-07-15 verification fetch round.

2. How fuel moves on Alaska winter roads (vehicles, loads, operators)

Three distinct operations exist, and they matter for cost:

(a) Engineered oilfield-standard ice roads (Deadhorse/Kuparuk/Alpine → Nuiqsut). A seasonal ice road of about 62 miles connects the Dalton Highway at Deadhorse to Nuiqsut (a ~31-mile segment ties Alpine/Kuparuk to the village) (dangerousroads.org; ADN 2014). These roads are built to oilfield spec and carry conventional highway tractors with tank trailers; upstream, Colville Inc. runs 54 tractors and 85 tank trailers hauling >2M gal/month up the Dalton to Prudhoe Bay (colvilleinc.com), and the same class of equipment makes the ice-road leg. Full or near-full tanker payloads (~5,000–8,000 gal) are feasible; speeds are ice-limited (roughly 15–25 mph loaded).

(b) Tundra packed-snow trails — the NSB Community Winter Access Trail (CWAT). First established winter 2017/2018 (ASTAR TM4), with BLM right-of-way authorized January 2019 (Alaska Business, verified), the CWAT is rebuilt annually from compacted snow: reconnaissance and pre-packing begin late December, trail construction runs through January, escorted caravans run mid-February through April, then the trail is decommissioned. Verified route lengths (Alaska Business): Prudhoe Bay–Utqiagvik 300 mi; **Utqiagvik–Atqasuk 68 mi**; Utqiagvik–Wainwright 127 mi; plus the southern route Galbraith Lake (Dalton Hwy)–Anaktuvuk Pass, the "109-mile route" of the 2026 haul. Before the CWAT, a historic winter trail on the same corridor carried Rolligon fuel hauls from Utqiagvik to Atqasuk (ASTAR 2019, verified — the study maps a distinct "Winter Fuel Delivery Route" but quantifies nothing about it). Haul vehicles are low-ground-pressure tracked tractors (PistenBully, Tucker; a Prinoth BR350 is also documented on the trail). In the **verified** Feb–Mar 2026 emergency operation, UIC Oil & Gas Support delivered ~100,000 gal to Anaktuvuk Pass in 12 runs (~8,333 gal/run) using PistenBullys towing fuel trailers, with a Sherp scouting river-ice crossings (~3 ft of water), six full-time crew (~10 with rotation), staged 147 mi from the Deadhorse OGS facility; early round trips took **7–10 days**, optimized to **~1.5 days** by late February (uicalaska.com, verified — no cost figures published). General CWAT usage 2022–23: ~58 caravans, ~140 vehicles, treks of 3–33 hours.

(c) River ice roads (Kuskokwim). The Kuskokwim Ice Road, maintained largely by the Native Village of Napaimute's crew with state funding, runs 200–300+ miles and connects 15+ villages (~11,000 people) to Bethel (ADN 2019; DOT&PF). Fuel moves two ways: vendors (Vitus Energy, Crowley from the Bethel terminal) run tanker trucks up-river when ice permits — in thin-ice years Vitus substituted fleets of smaller fuel trucks on routes scouted by Napaimute (tribalsafety.org) — and households self-haul drums/totes by pickup. Maintenance has cost \$300,000–\$370,000/season (KYUK; Alaska Native News).

Who pays what: road/trail provision is publicly funded — State of Alaska for the Kuskokwim; for the CWAT, a **verified FY2020 NPR-A impact grant of \$1,672,077** to the NSB (OMB project 62733). Hauling itself is done by borough enterprise operations or contractors (UIC OGS, Colville, Vitus, Crowley). NSB **residential** heating fuel is subsidized at \$1.50–2.50/gal depending on village (verified DCRA Winter 2026: AKP/Atqasuk/Nuiqsut \$1.50, Point Hope \$1.74, Wainwright \$1.75, Kaktovik \$2.50); the subsidy does not extend to businesses, so **commercial** prices must be used for revealed-cost inference. Utqiagvik reports no heating-oil price: homes heat with natural gas and the vendor declined to report a commercial price in Winter 2026.

3. Derivation log (steps and decisions)

- Searches 1–4** (Nuiqsut ice-road haul; Atkasuk/CWAT; Kuskokwim fuel trucking; NWT winter resupply rates). Established the three operational archetypes (§2). Found the UIC Anaktuvuk Pass 2026 haul — kept as the anchor real-world event. GNWT results gave program structure but no unit rates — kept for later differential use. *Rejected*: Ice Road Truckers/TV material and generic tanker-capacity marketing pages (no cost content).
- WebFetch attempts (4)** on UIC, Alaska Business CWAT, GNWT release, Alaska Beacon — all permission-denied at the time. Decision: proceed on snippets; flag PDFs for manual pull (later executed — see step 8).
- Searches 5–8** (Anaktuvuk haul cost; NSB budget "fuel haul"; DCRA prices for AKP/Atkasuk/Nuiqsut; Kuskokwim delivery savings). Got the key DCRA Winter 2026 commercial heating-oil prices. NSB budget snippets yielded only small "fuel home delivery" revenue lines (~\$59–86k, Atkasuk) — rejected as in-village delivery, not inter-community haul. Kuskokwim: delivered heating oil >\$6/gal region-wide vs. Bethel; no clean truck-rate found.
- Searches 9–12** (Tibbitt–Contwoyto rates; CWAT budget; Bethel/Aniak prices; Dalton trucking rates). Got Tibbitt scale figures; a secondary claim of "<C\$0.10/t-km" for winter-road trucking (Grokopedia — low reliability, kept only as a floor sanity check). Bethel heating oil \$3.50/gal (Vitus, Feb 2026) vs. Aniak \$6.22 (Crowley) — *rejected for Method B*: Aniak is served by its own summer barge, so the spread is not an ice-road haul price. Dalton per-gallon tanker rates: not published.
- Searches 13–16** (KYUK ice-road delivery pricing; Deadhorse retail; GNWT Déḻṉe/Colville Lake; Kuskokwim road funding). Kuskokwim road maintenance \$300k–\$370k/season confirmed; community plow service ~\$4,000/plow. GNWT: Déḻṉe gasoline C\$1.95/L after 2026 winter resupply; Colville Lake C\$2.44/L — kept for B3 corroboration. Deadhorse retail diesel: not found (one stale \$8.01 regular-gas datum — rejected).
- Searches 17–20** (ATRI trucking costs; AKP air-delivery costs; Canadian winter-road vs air rates; Nuiqsut route length). Got ATRI 2024 US average \$2.26/mi all-in truck operating cost (truckingresearch.org) — the backbone of Method A1. Got the Ontario Asheweig Winter Road manager quote (basis of B2). Got KNBA detail on AKP: Everts Air Cargo flies diesel weekly; one week's emergency airlift = 18,500 gal + 20 drums, plus 4,500 gal by ice road. Route lengths: Deadhorse–Nuiqsut ≈ 62 mi; CWAT to AKP = 109 mi.
- Stop rule reached (20 searches)**. Method weighting decided: A (build-up, two variants) + B (three price-differential variants) + C (Tibbitt tonne-km) + D (AKP event as emergency bound); weight community-scale ice-road tanker and snow-trail cat-train roughly equally; exclude industrial Tibbitt-scale economics from the point estimate (kept as floor).
- Verification round, 2026-07-15 (WebFetch restored)**. Confirmations and corrections:
- DCRA Winter 2026 (verified, full PDF, survey Jan 6–28 2026)**: full NSB table recovered — residential/commercial: AKP \$1.50/\$9.97, Atkasuk \$1.50/\$8.80, Kaktovik \$2.50/\$7.50, Nuiqsut \$1.50/\$7.70, Point Hope \$1.74/\$7.60, Utqiagvik n/a/n/a, Wainwright \$1.75/\$6.96. *Correction*: residential subsidy is \$1.50–2.50 by village, not \$1.50 everywhere. *New*: the marine-served commercial baselines (Kaktovik, Point Hope, Wainwright) enable a within-borough differential method — B1 rebuilt as B1' (see §4), retiring the Deadhorse-assumption chain.
- UIC AKP haul (verified)**: 100,000 gal / 12 runs / 109 mi confirmed. *Correction*: haul ran **Feb–Mar 2026**, not Jan–Apr. *New*: round-trip durations (7–10 days early → 1.5 days optimized), crew size (6 full-time, ~10 rotating), Sherp ice-profiling, 147-mi Deadhorse staging distance. No cost figures. A2 recalibrated on the observed 36-hr optimized round trip.
- Alaska Business CWAT (verified)**: Utqiagvik–Atkasuk leg is **68 mi** (was assumed ~60); other legs 300/127 mi; season calendar; ROW Jan 2019; 2022–23 usage stats.
- Northern Policy Institute (verified)**: *Correction*: article is **April 7, 2015** (Cheryl Reid), not ~2019 — B2 FX/CPI factors redone (2015 basis). Asheweig quote confirmed verbatim ("flying it in raises the cost to nearly \$2 a litre... winter roads lowers the price to \$1.35 a litre"); ~350 km serving six communities confirmed; hub price still not stated (remains an assumption). Context: Ontario's winter-road network is 3,160 km serving 30 First Nations + Moosonee; province contributed \$4.725M in 2011–12.
- Tibbitt–Contwoyto (verified via JV Geoscience Forum abstract, Burgundy, Wikipedia)**: *Correction*: annual operating budget is **C\$21–22M**, not C\$25M (the C\$25M snippet figure is retired). Yearly average 8,336 loads / 281,363 tons; ~400 km; >6,000 loads / 250,000 t typical; 2007 record 330,002 t / 10,922 loads; ~67-day season. Method C recomputed.
- GNWT (verified)**: full-cost-recovery pricing confirmed verbatim ("retail fuel prices at amounts that recover the full laid-in cost of fuel, including wholesale purchase and transportation costs"); June 2026 release names the 11 winter-resupply communities incl. Déḻṉe and Colville Lake; per-litre figures remain sourced from the Cabin Radio snippet.
- OMB FY2020 CWAT sheet (partially recovered)**: direct fetch cert-blocked; **\$1,672,077** NPR-A impact grant appropriation recovered from the search index of the same PDF (project 62733).
- ASTAR Atkasuk–Utqiagvik road study (verified, all 267 pages incl. TM4 and TM13)**: *Negative result*: the study contains **no** fuel-haul quantities or dollars anywhere — the Rolligon/CWAT fuel supply chain is described only qualitatively. Useful: CWAT origin winter 2017/2018; historic Rolligon fuel-haul route mapped; the all-season road alternative would cost **\$333–388M (\$5.2–5.4M/mi, 2019 Class 5)** — context for why the snow trail persists.
- Second pass, 2026-07-17 — four local primary PDFs read in full**. New findings and their effect:
- Noatak Winter Fuel Haul System (NWFHLS), 2012 legislative grant proposal (TPS 58673v1, DCCED Grants to Municipalities, Northwest Arctic Borough / Native Village of Noatak)**: The document the chapter was missing — and it corroborates Method A2's physics almost point-for-point, but supplies no operating cost. Quantitative content: **capital \$425,000** (\$350,000 for one *Arctic Snow Track Tractor + sled*, \$75,000 permitting/environmental/logistics planning) — matching the project's older \$425k research figure exactly. Route: a **20–30-mile overland winter road** from a DMTS-road (Red Dog Mine haul road) access point to Noatak, off-road tundra. Vehicle: single Arctic Snow Tractor, **10–12 mph, ~2-hr one-way trip** from any of three proposed corridors. **Payload 3,000–5,000 gal/trip**. Noatak annual fuel demand **350,000–400,000 gal/yr** (heating + power gen + vehicular); **~200 round trips/season** with one tractor; 3–5 seasonal jobs; system dormant in summer. Supply chain: fuel arrives at the Red Dog/DeLong Mountain Terminal (DMTS) port by ocean barge Jul–Sep, trucked up the DMTS road by a NANA-owned contractor (NANA Lynden), transferred to the snow tractor at a DMTS-road access point for the final 20–30-mi overland leg. Noatak currently gets **all** fuel by air (AVEC/school-district charters); retail fuel **>\$10/gal**, "highest cost of living in the nation." *Effect*: independently confirms A2's payload band (2,500–3,000 gal/unit → here 3,000–5,000 gal single sled), speed (5–12 mph tracked), and the single-tracked-unit architecture, and adds a second real Alaska route length (20–30 mi vs. AKP's 109 mi and Atkasuk's 68 mi). **But the proposal explicitly makes the haul a "self-sustaining business — operational costs will be paid for by the end-user" and gives NO operating**

\$/hr, \$/trip, \$/gal, or \$/gal-mi. So it does not replace A2's assumed \$900/hr convoy rate; it only tightens the physical inputs feeding it. A2's payload/speed assumptions are now corroborated by two independent Alaska sources; the convoy hourly rate remains the sole unverified load-bearing input.

19. **OMB FY2020 CWAT project sheet (AMD 62733) — now read directly (was cert-blocked in the 2026-07-15 round):** confirms the **\$1,672,077** FY2020 NPR-A Impact Grant appropriation, and reveals the CWAT program was awarded **\$4,000,000 total** (FY2019 supplemental + this FY2020 tranche), project dates 07/01/2019–06/30/2024. Project type "Life/Health/Safety"; scope is **maintaining preexisting trails and a Traveler Safety & Inspection Program in the NPR-A villages** — labor, fringe, contractual, equipment, materials, freight, travel, 25% admin. *Correction to the \$8 framing:* CWAT funding is **not** a fuel-haul subsidy — it funds trail maintenance and traveler safety, and the fuel haulers use the trail incidentally. The sheet lists **no trail mileage and no cost-per-mile**. So the \$8 trail-provision adder is re-derived on total-program dollars (\$4.0M over ~5 yr ≈ \$800k/yr) rather than the single-year \$1.67M, and the fuel-attribution share stays a modeling choice.
20. **NSB FY2026-2027 Budget Book (653 pp): No per-community gallons are published anywhere** — all fuel figures are dollar-denominated. Fuel operations live under Public Works → Enterprise & Energy Management → *Fuels & Natural Gas* (procurement, delivery, tank-farm O&M). Per-community fuel-operation budgets (FY26-27): Atkasuk 6108.FUEL **\$474,913** (3 FTE: fuel hazmat specialists + station attendants), AKP 6107.FUEL **\$401,222**, Nuiqsut 6110.FUEL **\$571,184** (+ separate Nuiqsut natural-gas op \$3,115,000), Point Lay 6112.FUEL **\$487,344**; Kaktovik / Point Hope / Anaktuvuk-6106 carry only a **\$2,000** placeholder (no staffed village fuel op). Per-community "Fuel Home Delivery" revenue (p.29): AKP \$86,873, Point Lay \$45,747, Atkasuk \$80,000, Nuiqsut \$12,426 (Nuiqsut also carries \$600,000 Commercial Gasoline revenue). Borough "Total Fuel" commodity line ≈ **\$9.8M** (general fund) plus **\$17.4M** power-generation fuel in the Enterprise Funds. **No CWAT / winter-trail budget line and no residential-subsidy total** appear in the FY26-27 book (the residential subsidy prices are in the DCRA report, §2). *Effect:* the budget confirms which villages run standalone fuel operations but cannot set gallon-weighted traffic; that comes from PCE (below).
21. **PCE FY2025 Statistical Report (per-community power-generation fuel):** supplies the per-community diesel **gallons** the budget lacks — these anchor the §7 traffic weighting. See §7 for the gallon-weighted mix and its effect on the point estimate.

Standing conversion factors (stated once, used throughout): CAD→USD 0.73 (2026), 0.78 (2015); US CPI inflator 2015→2026 ×1.36, 2021→2026 ×1.20, 2022→2026 ×1.11; arctic diesel/heating oil density 0.84 kg/L → 1 tonne ≈ 1,190 L ≈ 313 US gal; 1 km = 0.621 mi; 1 gal = 3.785 L.

4. Method A / B / C — full derivations with arithmetic

Method A — engineering cost build-up (two route variants)

A1. Ice-road tanker (Nuiqsut / Kuskokwim type). - Base all-in truck cost: ATRI 2024 = \$2.26/mi at ~40 mph effective → $2.26 \times 40 = \text{\$90/hr}$ (lower-48, 2024\$). - Alaska remote-winter premium ×1.75 (short season → poor utilization, mobilization, winterization, higher wages, standby): $90 \times 1.75 \approx \text{\$158/hr} \approx \text{\$160/hr}$ (2026\$; ATRI 2024→2026 inflation is inside the 1.75 rounding). - Ice-road speeds: 15 mph loaded / 25 mph empty. Time per delivered mile (round trip) = $1/15 + 1/25 = 0.1067$ hr/mi → $160 \times 0.1067 = \text{\$17.1/delivered-mile}$. Add 20% for loading/unloading/convoy delays → **\$20.5/delivered-mile**. - Payload: 5,000 gal typical community tanker (range 2,500 gal thin-ice small trucks → 7,500 gal full trailer on oilfield-spec ice). - Rate = $20.5 / 5,000 = \text{\$0.0041/gal-mi}$ (range **\$0.0027–0.0082**).

A2. Tundra snow-trail cat-train (Atkasuk / Anaktuvuk type), calibrated to the **verified 2026 AKP haul** (100,000 gal, 12 runs, ~8,333 gal/run, 109 mi, PistenBullys + fuel trailers, 6 crew). *Physical inputs now independently corroborated (2026-07-17) by the 2012 Noatak NWFHLS proposal: single Arctic Snow Tractor + sled, 3,000–5,000 gal/trip, 10–12 mph, ~200 round trips to move 350–400k gal/yr over a 20–30-mi overland leg — the same payload/speed/single-unit architecture assumed here. Noatak states no operating cost, so the calibration below is unchanged.* - Convoy cost: ~3 tracked units with operators @ ~\$275/hr (typical Alaska oilfield tracked-support rates, **assumption — still the only unverified input**) + \$75/hr supervision/support = **\$900/hr**. *Sanity check from Noatak: a one-tractor, 3–5-job, self-sustaining operation implies a crew/equipment hourly well below the 3-unit AKP convoy; the AKP-derived \$900/hr therefore sits at the emergency/multi-unit end of the range, consistent with treating the routine rate as ~60% of it (below).* - Trip time: **verified** — optimized round trips of ~1.5 days (**36 hr**) by late February. (The original blind assumption was 32.5 hr from 7/10 mph speeds + 6 hr load/rest; the observed 36 hr replaces it — a 10% correction.) Early trail-breaking round trips took 7–10 days (2 ft of new snow); even with idle time billed at reduced standby rates these runs cost roughly 2× per gallon, so the campaign average sits above the optimized figure. - Optimized run: $900 \times 36 \approx \text{\$32,400/run} \rightarrow 32,400 / 8,333 \text{ gal} \approx \text{\$3.89/gal over 109 mi} \rightarrow \text{\$0.036/gal-mi}$ (emergency-campaign conditions, established trail). Campaign floor $12 \times \$32.4k \approx \$390k$. Cross-check: airlifting 100,000 gal at typical flown-fuel freight of \$3.5–5/gal would cost \$350–500k, and the verified weekly airlift ceiling was ~18,500 gal — the Borough's overland choice is consistent with roughly airlift-parity cost at much higher weekly capacity. - Routine, planned haul (better utilization, no standby/urgency premium in the hourly rate): ~60% of emergency cost → **≈\$0.021/gal-mi** (range \$0.017–0.036, upper end = emergency).

Method B — revealed retail price differentials

B2 (primary). Ontario Asheweig winter-road system (~350 km serving six First Nations; Northern Policy Institute, **April 2015, verified**): gasoline flown in ≈ C\$2.00/L; via winter road ≈ C\$1.35/L (quote verified verbatim). Road-served hub price at the time ≈ C\$1.05–1.15/L (**assumption** — the article does not state it; Pickle Lake–area). Winter-road haul + local increment ≈ C\$0.20–0.30/L, take **C\$0.25/L** over an average ~250 km (155 mi) haul: $C\$0.25/L \times 3.785 \text{ L/gal} = C\$0.946/\text{gal} \div 155 \text{ mi} = C\$0.0061/\text{gal-mi}$ (2015 C\$) × 0.78 USD/CAD = US\$0.0048 × 1.36 (CPI 2015→2026) = **US\$0.0065/gal-mi** — and this still contains some local vendor margin, so pure haul ≈ **\$0.004–0.007/gal-mi**.

B3 (corroboration). GNWT Délı̨nę (2026 winter-resupply-priced gasoline C\$1.95/L; full-cost-recovery pricing **verified verbatim**: prices "recover the full laid-in cost of fuel, including wholesale purchase and transportation costs") vs. Yellowknife-area rack ≈ C\$1.35/L: differential C\$0.60/L over ~800 km effective haul (highway + Mackenzie Valley winter road): $C\$0.60 \times 3.785 = C\$2.27/\text{gal} \div (800 \times 0.621 = 497 \text{ mi}) = C\$0.00457/\text{gal-mi} \times 0.73 = \text{\$0.0033/gal-mi}$ — includes GNWT storage/handling, so pure haul is somewhat lower; consistent with B2 given the longer, partly-highway route.

B1' (rebuilt on the verified DCRA table — within-borough differentials). All NSB villages carry comparable small-market tank-farm/utility overhead and the same commercial (unsubsidized) pricing regime, so comparing winter-road-served villages against the Borough's **marine-served commercial baseline (\$6.96–7.60**: Wainwright \$6.96, Kaktovik \$7.50, Point Hope \$7.60) nets out most of the non-transport overhead that contaminated the original B1: - **Nuiqsut \$7.70** (engineered ice road, 62 mi from Deadhorse): premium \$0.10–0.74 → **\$0.002–0.012/gal-mi gross** — Nuiqsut prices like a marine-served village; the ice-road tanker leg adds almost nothing. Corroborates A1. - **Atkasuk \$8.80** (snow trail, 68 mi from Utqiagvik): premium \$1.20–1.84 → **\$0.018–0.027/gal-mi gross** — lands squarely on the A2 routine band. - **AKP \$9.97** (109-mi snow trail + flown fuel): premium \$2.37–3.01 → **\$0.022–0.028/gal-mi gross** (a snow-trail/airlift blend). Caveats: single-vendor point-in-time prices, and the upstream supply basis differs (Deadhorse trucked fuel vs. ocean barge), so these stay gross corroborations rather than point inputs. *The original B1 (Deadhorse laid-in assumption chain → \$0.063 gross) is retired: its \$3.90 "differential" was mostly tank-farm and utility overhead that the within-borough comparison removes.*

Method C — circumpolar heavy-haul tonne-km economics (Tibbitt–Contwoyto)

Verified inputs: annual operating budget **C\$21–22M** (JV, via NWT Geoscience Forum); yearly average **8,336 loads / 281,363 tons**; ~400 km; season ~67 days. - Road provision: $C\$21.5M \div 281,363 \text{ t} \approx C\$76/\text{t}$; average haul $\approx 350 \text{ km} \rightarrow$ **C\$0.22/t-km** road share. - Line-haul trucking on top: $C\$0.15 - 0.25/\text{t-km}$ (ice-speed truckload work) → all-in $\approx$ **C\$0.37–0.47/t-km**; adopt $C\$0.42/\text{t-km}$ central. (The " $<C\$0.10/\text{t-km}$ " claim found for winter-road trucking evidently excludes road provision; rejected as a central value, retained as a floor.) - Convert for diesel: $C\$0.42 \div (313 \text{ gal/t} \times 0.621 \text{ mi/km}) = C\$0.42/194.4 = C\$0.00216/\text{gal-mi} \times 0.73 =$ **US\$0.0016/gal-mi** (range $US\$0.0014 - 0.0018$; floor from the $C\$0.10$ claim $\approx US\$0.0004$). (Pre-verification value was \$0.0017 off the C\$25M snippet — a 6% shift.) - Interpretation: this is the **industrial-scale floor** — 6,000–10,000 loads amortizing one road. Community hauls (tens of loads) cannot reach this efficiency.

Method D — documented event bound (emergency)

The **verified** Feb–Mar 2026 Anaktuvuk Pass operation (Method A2's calibration event) implies $\approx$ **\$0.036/gal-mi under emergency conditions on an established trail** (36-hr round trips), and materially more during the trail-breaking weeks (7–10-day round trips); the airlift alternative it supplemented (Everts Air Cargo, 18,500 gal flown in a week) sits at roughly \$0.03–0.05/gal-mi-equivalent. **Labeled emergency bound, not routine.**

5. Source table

Source	URL	Year	What it provided	Reliability
DCRA Alaska Fuel Price Report, Winter 2026 (PDF)	commerce.alaska.gov/.../Alaska%20Fuel%20Price%20Report%20-%20January%202026.pdf	2026	Verified in full. NSB residential/commercial table: AKP \$1.50/\$9.97, Atkasuk \$1.50/\$8.80, Kaktovik \$2.50/\$7.50, Nuiqsut \$1.50/\$7.70, Pt Hope \$1.74/\$7.60, Wainwright \$1.75/\$6.96, Utqiagvik n/a (gas; vendor declined commercial). Survey Jan 6–28, 2026; 92-community unsubsidized avg \$6.49	High (verified 2026-07-15)
UIC Alaska — AKP emergency fuel haul	uicalaska.com/2026/04/15/uicogs-emergency-fuel-haul-to-anaktuvuk-pass/	2026	Verified. ~100,000 gal, 12 runs, 109 mi, Feb–Mar 2026; PistenBullys + fuel trailers, Sherp scout; 6 crew (~10 rotating); round trips 7–10 d → 1.5 d; staging 147 mi from Deadhorse. No cost figures	High (verified; operator's own account)
KNBA — AKP fuel shortage	knba.org/news/2026-01-15/...	2026	Everts Air Cargo weekly fuel flights; 18,500 gal airlift week; 4,500 gal by ice road	High (snippet only — not re-fetched)
Alaska Business — "The Community Winter Access Trail"	akbizmag.com/industry/transportation/cwat/	2024	Verified. Route lengths: Prudhoe–Utqiagvik 300 mi, Utqiagvik–Atkasuk 68 mi , Utqiagvik–Wainwright 127 mi; season calendar (recon Dec, build Jan, caravans mid-Feb–Apr); ROW Jan 2019; 2022–23: 58 caravans/140 vehicles; equipment (PistenBully 600, Tucker, Prinoth BR350). No haul-cost figures	Med-High (verified)
ASTAR — Atkasuk to Utqiagvik All Season Access Road (ASRC Energy Services for ADNR, draft Rev. 1)	north-slope.org/.../ASTAR-ATQ-to-UTQ-Study.pdf	2019	Verified, all 267 pp incl. TM4/TM13. CWAT established winter 2017/2018; historic Rolligon fuel-haul route mapped (Fig. TM4-5); road alternative \$333–388M (\$4.8M/mi roadway, \$5.2–5.4M/mi avg, Class 5, 2019\$). Contains zero fuel-haul quantities or dollars	High (verified; negative result for haul cost)
BLM blog — CWAT	blm.gov/blog/2019-11-22/...	2019	CWAT purpose, cost/risk reduction	Medium
ADN / dangerousroads — Nuiqsut ice road	adn.com/.../plan-permanent-road...; dangerousroads.org/.../6936-nuiqsut-ice-road.html	2014/undated	Route lengths: ~62 mi Deadhorse–Nuiqsut; ~31 mi Alpine segment	Medium
KYUK — Kuskokwim ice road state funding	kyuk.org/politics/2021-07-06/...	2021	\$370k season budget; Napaimute crew; \$4,000/plow	High
Alaska DOT&PF press release	dot.alaska.gov/comm/pressbox/arch2022/PR22-0010.shtml	2022	State funding; >\$6/gal delivered heating oil regionally; 300-mi road, 17 villages	High
tribalsafety.org — GPR on Alaska ice roads	tribalsafety.org/ground-penetrating-radar-...	~2020	Vitus small-truck fleet substitution on thin ice	Medium
Colville Inc. — Transport	colvilleinc.com/our-services/transport/	current	54 tractors / 85 tank trailers; >2M gal/month up the Dalton	High

Source	URL	Year	What it provided	Reliability
ATRI Operational Costs of Trucking 2025 update	truckingresearch.org/2025/07/...	2025 (2024 data)	\$2.26/mi all-in US truck operating cost	High
Northern Policy Institute — "Weathering Winter Roads – What is the Best Route?" (Cheryl Reid)	northernpolicy.ca/article/weathering-winter-roads-...-1354.asp	2015 (verified — was misdated ~2019)	Verified verbatim: Asheweik flown gas ≈ C\$2.00/L vs C\$1.35/L via winter road; ~350 km / six communities; Ontario network 3,160 km / 30 First Nations; province \$4.725M (2011–12); all-weather road \$1.6–3.0M/km	Medium (verified)
GNWT newsroom — fuel prices updated, 11 NWT communities	gov.nt.ca/en/newsroom/fuel-prices-updated-11-nwt-communities (+ Cabin Radio)	2026	Verified: full-cost-recovery statute language; 11 winter-resupply communities named (incl. Délı̨nę, Colville Lake). Per-litre figures (Délı̨nę C\$1.95, Colville Lake C\$2.44) remain from Cabin Radio snippet	High (release verified; prices snippet-level)
Tibbitt–Contwoyto JV — NWT Geoscience Forum abstract; jvtcwinterroad.ca; Burgundy Diamonds; Wikipedia	nwtgeoscience.ca/gsforum/...; jvtcwinterroad.ca; burgundydiamonds.com; en.wikipedia.org/wiki/Tibbitt_to_Contwoyto_Winter_Road	current	Verified: annual operating budget C\$21–22M (corrects C\$25M snippet); avg 8,336 loads / 281,363 tons; ~400 km, 85% over frozen lakes; >6,000 loads / 250,000 t typical; 2007 record 330,002 t / 10,922 loads; ~67-day season	Med-High (verified)
Grokopedia — winter road	grokopedia.com/page/Winter_road	current	"<C\$0.10/t-km" trucking claim; air 6× multiplier	Low (floor check only)
DCRA Bethel/Aniak prices (state heating-oil sheet)	oppm.doa.alaska.gov/media/1337/08-heating-oil.pdf	2026	Bethel \$3.50 (Vitus), Aniak \$6.22 (Crowley)	Medium (rejected for method use)
Alaska OMB FY2020 CWAT capital project sheet (proj. 62733 / AMD 62733)	omb.alaska.gov/ombfiles/20_budget/DCCED/Enacted/2020proj62733.pdf (local PDF)	2019	Read in full 2026-07-17. FY2020 appropriation \$1,672,077 confirmed; CWAT total award \$4,000,000 (FY2019 supp + FY2020); dates 07/01/2019–06/30/2024; scope = trail maintenance + Traveler Safety & Inspection Program in NPR-A villages (Life/Health/Safety); no trail mileage, no cost-per-mile, no fuel-haul language	High (verified)
Northwest Arctic Borough — Noatak Winter Fuel Haul System (NWFHLS) legislative grant proposal (TPS 58673v1, DCCED Grants to Municipalities AS 37.05.315)	local PDF (2012 Legislature Total Project Snapshot Report)	2012	Read in full 2026-07-17. The chapter's missing overland-haul engineering source. Capital \$425,000 (1 Arctic Snow Track Tractor + sled \$350k; permitting/planning \$75k); 20–30-mi overland winter road off the DMTS road; 10–12 mph, ~2-hr one-way; 3,000–5,000 gal/trip ; Noatak demand 350–400k gal/yr, ~200 round trips/season , 3–5 jobs; supply via Red Dog/DMTS barge→truck→snow-tractor. Corroborates A2 payload/speed/architecture; states NO operating \$/gal or \$/gal-mi ("self-sustaining, end-user pays operating cost")	Med-High (official grant doc; proposal, not an executed cost study)
North Slope Borough FY2026-2027 Budget Book (653 pp)	local PDF (nsb.gov)	2025	Read (targeted) 2026-07-17. No per-community gallons. Per-community fuel-op budgets FY26-27: Atkasuk \$474,913, AKP \$401,222, Nuiqsut \$571,184 (+\$3.115M Nuiqsut gas), Point Lay \$487,344; Kaktovik/Pt Hope/AKP-6106 = \$2,000 placeholder. "Fuel Home Delivery" revenue: AKP \$86,873, Pt Lay \$45,747, Atkasuk \$80,000, Nuiqsut \$12,426. Borough Total Fuel ≈ \$9.8M general + \$17.4M power-gen. No CWAT line; no residential-subsidy total	High (primary budget)
Alaska PCE FY2025 Statistical Report by Community	local PDF (AEA, reporting period 07/01/24–06/30/25)	2026	Read 2026-07-17. Per-community power-generation diesel gallons — sets the \$7 gallon-weighted tanker-vs-sled traffic mix	High (state statutory report)

6. Route-type differences

Costs differ systematically by roughly an order of magnitude across route types:

Route type	Vehicle / payload	Speed / tempo	2026 USD \$/gal-mi
Industrial engineered ice road (Tibbitt-scale, thousands of loads)	Highway tanker, full loads	15–25 mph	0.001–0.002 (floor)
Community engineered/river ice road (Nuiqsut, Kuskokwim)	Tanker truck, 2,500–7,500 gal	15–25 mph	0.003–0.008
Tundra packed-snow trail (CWAT: Atkasuk 68 mi, AKP 109 mi)	Tracked tractor + fuel trailers, ~2,500–3,000 gal/unit	5–10 mph; 36-hr optimized round trip on the 109-mi AKP run (verified)	0.017–0.036 (upper end = emergency)

The driver is payload × speed: a cat-train moves roughly one-third the fuel at one-third the speed of an ice-road tanker, for a similar-or-higher hourly cost. The verified within-borough retail premiums (\$4 B1') reproduce the same order-of-magnitude split: Nuiqsut +\$0.10–0.74 over marine-served villages vs. Atkasuk +\$1.20–1.84 and AKP +\$2.37–3.01.

7. Cross-method comparison and reconciliation

- **A1 (build-up, ice-road tanker): \$0.0041** (0.0027–0.0082) — independent of any price data.
- **B1' (verified within-NSB differential, Nuiqsut): \$0.002–0.012 gross** — an ice-road-served village prices almost identically to marine-served villages; converges with A1.
- **B2 (Ontario revealed differential, corrected to 2015 basis): \$0.0065**, with margin contamination pushing the pure-haul figure toward **\$0.004–0.007** — converges with A1.
- **B3 (NWT Déline): \$0.0033** — consistent, given partial highway routing.
- **C (Tibbitt tonne-km, verified budget): \$0.0014–0.0018**, central \$0.0016 — deliberately below the community band; explains itself via scale (8,000+ loads vs. dozens), so no conflict.
- **A2/D (snow trail, recalibrated on the verified 36-hr round trip): \$0.017–0.036**, routine $\approx$ **\$0.021** — an order above the tanker band, and now **directly corroborated** by the verified within-borough premiums (Atkasuk \$0.018–0.027, AKP \$0.022–0.028 gross), which land inside the A2 band rather than merely bounding it.

Three genuinely independent families (engineering build-up; revealed price differentials in two countries — now including a clean within-borough comparison; heavy-haul tonne-km economics) bracket the community ice-road tanker cost at **\$0.003–0.008/gal-mi** and the snow-trail cat-train at **\$0.017–0.03/gal-mi**. A single mode rate must sit between them, weighted by the gallon traffic split between the tanker-served (engineered ice road) and sled-served (snow trail) regimes.

Traffic weighting — now anchored on PCE gallons (2026-07-17), previously an equal-weight assumption. The two Alaska communities whose fuel actually moves on winter overland routes in the model's `ice_road` mode are **Nuiqsut** (engineered ice road, tanker regime) and **Atkasuk** (packed-snow trail, sled regime). The NSB budget publishes no gallons, but the PCE FY2025 report gives per-community power-generation diesel: **Nuiqsut 279,232 gal vs. Atkasuk 274,877 gal — essentially 50/50**. Total road-delivered throughput (heating + vehicle + power) tilts slightly toward the tanker side: Nuiqsut's population (500 vs. 307), NSB fuel-operation budget (\$571k vs. \$475k), and larger tank farm all point to modestly more gallons on the ice-road/tanker regime, partly offset by Nuiqsut having shifted much of its power generation to natural gas (so less *liquid* fuel per capita). Anaktuvuk Pass (320,380 gal) is deliberately **excluded from the weighting** — despite the dramatic 2026 emergency overland haul, AKP is resupplied primarily by air, not by a standing winter road, so its gallons are not sled-regime traffic (they remain the \$4-D emergency bound). Net: the real gallon-weighted split is **~50/50, tilting to perhaps 55/45 in favor of the cheaper tanker regime** — which *confirms* the prior equal-weight assumption rather than moving it, and if anything nudges the point estimate very slightly downward.

- Equal weight (50/50): arithmetic mean of 0.0045 and 0.021 $\approx$ **0.013**; geometric mean $\approx \sqrt{0.0045 \times 0.021} \approx$ **0.0097**.
- Tanker-tilted (55/45): arithmetic $\approx 0.055 \times 0.021$ rescaled $= 0.45 \times 0.021 + 0.55 \times 0.0045 \approx$ **0.0119**; geometric $\approx 0.0045^{0.55} \times 0.021^{0.45} \approx$ **0.0092**.

The report takes **\$0.010/gal-mi**, leaning geometric because the high-end (sled) figure carries residual emergency contamination. The PCE-anchored weights leave both means straddling 0.010, so the point estimate is **unchanged**; what changed is its status — the tanker-vs-sled mix is now data-backed (§9 item 2 resolved) rather than a coin-flip, raising confidence in the point value from low-to-moderate to moderate. (Pre-verification means were 0.012/0.0092 $\rightarrow$ same rounded point value.)

8. Final estimate: ONE recommended rate

Recommended rate: \$0.010 per gallon-mile (2026 USD) — unchanged by the verification round. - **Range:** \$0.004 (engineered ice road, full tanker loads, routine season) to \$0.027 (tundra snow-trail cat-train, difficult conditions). Absolute floor for industrial-scale operations: $\sim$ \$0.001–0.002. Emergency snow-trail operations run $\approx$ \$0.036 (verified-calibrated) and are excluded from the range. - **Confidence:** Moderate-to-high in the range — both sub-bands are now corroborated by verified within-borough price differentials independent of the engineering assumptions, and A2's payload/speed/architecture are independently confirmed by the 2012 Noatak NWFHLS proposal. **Raised to moderate in the point value** (was low-to-moderate): the tanker-vs-sled traffic mix is now anchored on PCE per-community gallons (\$7, $\sim$ 50/50) instead of an assumption. A2's $\sim$ \$900/hr convoy rate remains the one material unverified input — the Noatak proposal supplies the physics but no operating cost. - **Includes:** vehicle capital and operating cost, crew, fuel burned, empty backhaul, loading/unloading, contractor overhead and margin, and the hauler's incidental trail-packing during the haul. - **Excludes:** the fuel commodity itself; upstream line-haul to the winter-road origin (Dalton Highway trucking, ocean/river barge to Utqiagvik or Bethel); publicly funded ice-road/trail construction and maintenance. *Trail-provision adder, re-derived on the now-read OMB sheet:* CWAT is a **\$4.0M total** award (FY2019 supp + FY2020 \$1,672,077) over 07/2019–06/2024 $\approx$ **\$800k/yr** for the $\sim$ 500–600-trail-mile NPR-A network, and the sheet's scope is **trail maintenance + traveler safety, not fuel-haul funding** (haulers use the trail incidentally). So if the model must internalize trail provision, apply a 10–30% fuel-traffic attribution to $\approx$ \$800k/yr, adding roughly **\$0.001–0.003/gal-mi** (unchanged magnitude, now on firmer footing; Kuskokwim runs \$300–370k/season). Also excluded: tank-farm storage and retail markup; and emergency premiums ($\approx$ \$0.036 overland; displaced airlift $\sim$ \$0.03–0.05).

9. What would most improve this estimate

(Resolved 2026-07-15: the six flagged sources were fetched — see §3 step 8. Resolved 2026-07-17: four local primary documents read — see §3 step 9 and §10. Item 2 (traffic weights) and item 4 (OMB sheet text) below are now closed; item 1 remains the single top gap.)

Remaining gaps, in order of leverage: 1. **An actual invoice or contract rate** for one haul — now unambiguously the top and only material gap. Neither the 2026-07-15 fetch round nor the 2026-07-17 local documents state one: the Noatak proposal made the haul a "self-sustaining business, end-user pays operating cost" and quoted no rate; the NSB budget publishes fuel operations only in aggregate dollars, not per-haul. The NSB Public Works / UIC OGS contract for the Atkasuk or AKP fuel haul (a FOIA-able borough expenditure), Vitus/Crowley's per-gallon Kuskokwim truck surcharge, or the executed Noatak NWFHLS operating budget would each replace A2's assumed $\sim$ \$900/hr convoy rate — the last load-bearing assumption — with a hard number. 2.

~~**Gallon-mile traffic weights** between tanker- and sled-served routes~~ — **RESOLVED 2026-07-17**. PCE FY2025 per-community diesel gallons (Nuiqsut 279,232 / Atkasuk 274,877, ~50/50) now anchor the §7 weighting; the point estimate held at \$0.010 and its confidence rose to moderate. 3. **Deadhorse laid-in fuel cost** (Colville wholesale), which would let the Nuiqsut differential be decomposed into supply-basis vs. haul components instead of relying on the marine-served baseline. 4. ~~**OMB project sheet text**~~ — **RESOLVED 2026-07-17**. Read in full: \$4.0M total CWAT award, trail-maintenance/traveler-safety scope (not fuel-haul funding), no trail mileage or cost-per-mile. §8 trail-provision adder re-derived accordingly; magnitude unchanged (~\$0.001–0.003/gal-mi). 5. **A per-community total-fuel gallons series (heating + vehicle + power)** would refine the §7 weights beyond the power-gen-diesel proxy used here — heating oil is the bulk of village throughput and is not captured by PCE generation gallons. Low leverage (would shift weights within the ~50/55% band), but the natural next increment.

10. Verification addendum — second pass (local primary documents, 2026-07-17)

Four downloaded primary PDFs were read to test whether any contained the winter fuel-haul cost data the chapter had been missing. Summary of what each contributed and the net effect on the estimate.

(1) Northwest Arctic Borough — Noatak Winter Fuel Haul System (NWFHLS), 2012 legislative grant proposal (TPS Report 58673v1) — the headline document. This is a real Alaska overland-fuel-haul engineering proposal, and it is the closest primary source to the chapter's Method A2. What it gives: - **Capital: \$425,000** — \$350,000 to procure one *Arctic Snow Track Tractor + sled* "capable of hauling fuel off road for up to 30 miles in winter conditions," \$75,000 for permitting/environmental/logistics planning. (Matches the project's older \$425k research figure exactly, and independently confirms the ~\$425k Arctic-tractor capital cost.) - **Route: a 20–30-mile overland (tundra) winter road** from an access point on the DMTS/Red Dog haul road to Noatak; three candidate corridors. - **Vehicle performance:** single Arctic Snow Tractor, **10–12 mph, ~2-hr one-way trip, 3,000–5,000 gal/trip**. - **System scale:** Noatak annual fuel demand **350,000–400,000 gal/yr** (heating + power + vehicular); **~200 round trips/season** with one tractor; **3–5 seasonal jobs**; dormant in summer. - **Supply chain:** barge to Red Dog/DMTS port (Jul–Sep) → NANA-Lynden truck up the DMTS road → transfer to snow tractor for the final 20–30-mi overland leg. - **Baseline it displaces:** Noatak is 100% air-served (AVEC/school charters), retail fuel **>\$10/gal**, "highest cost of living in the nation."

Critical negative result: the proposal explicitly structures the haul as a "self-sustaining business — operational costs will be paid for by the end-user" and gives **no operating cost per hour, per trip, per gallon, or per gallon-mile**, and no projected \$/gal savings figure (the "savings" case is qualitative: economies of scale + money staying in the community). So it **corroborates every physical input to Method A2** (payload band, tracked speed, single-unit architecture, trips-per-season order of magnitude) and adds a third real Alaska route length (20–30 mi, alongside AKP's 109 mi and Atkasuk's 68 mi), **but it does not replace A2's assumed ~\$900/hr convoy rate**. That assumption remains the single load-bearing unverified input in the whole chapter.

(2) OMB FY2020 CWAT project sheet (AMD 62733) — now read directly (was certificate-blocked in the 2026-07-15 round; only the appropriation amount had been recovered from the PDF search index). Confirms **\$1,672,077** FY2020 NPR-A Impact Grant; reveals the CWAT program's **total award of \$4,000,000** (FY2019 supplemental + this FY2020 tranche), project dates 07/01/2019–06/30/2024, type Life/Health/Safety, scope = **maintaining preexisting trails + a Traveler Safety & Inspection Program** in NPR-A villages. **No trail mileage, no cost-per-mile, no fuel-haul language.** Effect: the §8 trail-provision adder is re-based on ~\$800k/yr (\$4.0M / 5 yr) and reframed as *not* a fuel-haul subsidy; adder magnitude unchanged (~\$0.001–0.003/gal-mi).

(3) North Slope Borough FY2026-2027 Budget Book (653 pp). No per-community gallons anywhere — fuel is dollar-denominated throughout. Useful lines: per-community fuel-operation budgets (Atkasuk \$474,913, AKP \$401,222, Nuiqsut \$571,184 + \$3.115M Nuiqsut gas, Point Lay \$487,344; Kaktovik/Point Hope/AKP-6106 only a \$2,000 placeholder — i.e., these three run no staffed village fuel op); "Fuel Home Delivery" revenue (AKP \$86,873, Point Lay \$45,747, Atkasuk \$80,000, Nuiqsut \$12,426); borough Total Fuel ≈ \$9.8M general fund + \$17.4M power-gen. No CWAT budget line and no residential-subsidy total appear in the FY26-27 book. Effect: confirms which villages operate standalone fuel divisions but cannot set gallon weights — that comes from PCE.

(4) Alaska PCE FY2025 Statistical Report — supplies the per-community gallons the budget lacks. Power-generation diesel (FY2025): Nuiqsut 279,232 gal, Atkasuk 274,877 gal, Anaktuvuk Pass 320,380, Kaktovik 381,788, Point Hope 556,245, Point Lay 297,685, Wainwright 548,000, Noatak 134,327. The two winter-overland-served archetype communities (Nuiqsut tanker / Atkasuk sled) are within 2% of each other on power-gen diesel — a **~50/50 traffic split**, which anchors the §7 weighting and *confirms* the prior equal-weight assumption. Caveat: power-gen diesel omits heating oil (the bulk of throughput), so this is a proxy; total-fuel gallons would refine but not overturn the weight (see §9 item 5).

Net effect on the recommended rate: UNCHANGED at \$0.010/gal-mi (range \$0.004–0.027 per §8; emergency ≈ \$0.036 excluded). The Noatak document, despite being the most promising lead, contained no operating cost, so Method A2's build-up stands as-is; the PCE gallons confirmed rather than moved the §7 weighting. What improved is confidence in the **point value** (low-to-moderate → moderate), because the tanker-vs-sled traffic mix is now data-backed. The estimate's one remaining soft spot is A2's ~\$900/hr convoy rate; closing it now requires an actual haul invoice/contract (§9 item 1), which no public document in either verification round has provided.